

The genetic and dietary landscape of the muscle insulin signalling network

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 Copyright information

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

January 10, 2024 (this version)

Reviewed preprint version 1

September 12, 2023

Posted to preprint server

June 22, 2023

Sent for peer review

June 6, 2023

Abstract

Metabolic disease is caused by a combination of genetic and environmental factors, yet few studies have examined how these factors influence signal transduction, a key mediator of metabolism. Using mass spectrometry-based phosphoproteomics, we quantified 23,126 phosphosites in skeletal muscle of five genetically distinct mouse strains in two dietary environments, with and without acute *in vivo* insulin stimulation. Almost half of the insulin-regulated phosphoproteome was modified by genetic background on an ordinary diet, and high-fat high-sugar feeding affected insulin signalling in a strain-dependent manner. Our data revealed coregulated subnetworks within the insulin signalling pathway, expanding our understanding of the pathway's organisation. Furthermore, associating diverse signalling responses with insulin-stimulated glucose uptake uncovered regulators of muscle insulin responsiveness, including the regulatory phosphosite S469 on Pfkfb2, a key activator of glycolysis. Finally, we confirmed the role of glycolysis in modulating insulin action in insulin resistance. Our results underscore the significance of genetics in shaping global signalling responses and their adaptability to environmental changes, emphasizing the utility of studying biological diversity with phosphoproteomics to discover key regulatory mechanisms of complex traits.

eLife assessment

This **fundamental** study provides a unique tool for assessing the range of phosphorylation in insulin reactions due to genetic variation and dietary influence through the utilization of genetically distinct mouse strains. The discoveries of this study hold substantial importance, as they shed light on the interplay between genetic attributes and environmental conditions in shaping the insulin-signaling network within skeletal muscle, a crucial regulator of metabolism. The supporting evidence presented is **compelling**, and the work is anticipated to captivate a wide audience within the metabolism discipline due to its extensive appeal and by providing inspiration for further hypothesis-driven research.

Introduction

Protein post-translational modifications such as phosphorylation enable cells to rapidly respond to environmental changes by modifying protein function at low metabolic cost¹. As a result of this high metabolic efficiency, phosphorylation is involved in nearly all biological processes and is dysregulated in numerous complex diseases². Advances in mass spectrometry-based phosphoproteomics – the unbiased identification and quantification of protein phosphorylation – have led to the discovery of more than 100,000 phosphosites, revealing that the phosphoproteome comprises vast, interconnected phosphorylation networks^{3–6}, rather than the textbook view of isolated, linear kinase cascades.

The insulin signalling network is among the most studied phosphorylation networks. Post-prandial increases in blood glucose stimulate pancreatic insulin secretion, coordinating a metabolic switch in target tissues like skeletal muscle and adipose⁷. Insulin increases glucose uptake into these tissues by promoting translocation of the glucose transporter GLUT4 to the plasma membrane, and serves other functions like enhancing protein synthesis, downregulating lipid catabolism, and altering gene transcription⁷. To coordinate these functions, insulin triggers a phosphorylation cascade primarily involving activation of the Ser/Thr kinase Akt, regulation of downstream kinases including mTORC1 and GSK3, and modulation of parallel signalling arms^{7–9}. Seminal phosphoproteomics studies demonstrated that this cascade regulates over a thousand phosphosites, with many still uncharacterised in insulin action^{10–12}. Insulin resistance – the failure of insulin to promote glucose uptake in its target tissues – is triggered by genetic and environmental factors such as family history of metabolic disease and high-calorie diets¹³. Although insulin resistance is a major precursor of metabolic disease including type 2 diabetes, its mechanistic basis remains unresolved^{13–15}.

Interactions between genetics and environment significantly regulate biomolecular processes, including insulin resistance^{16–19}. As signalling pathways connect the extracellular environment to intracellular proteins, they are likely a major conduit of gene-by-environment interactions. Yet, how global phosphorylation signalling networks are regulated across different genetic backgrounds is relatively unexplored. Recent phosphoproteomics studies in yeast²⁰ and mice²¹ identified genetic variants affecting multiple phosphosites, but did not analyse the phosphoproteome's response to acute perturbation, which is crucial to its role as a signal transduction system. We have also shown marked variation in acute signalling responses to exercise or insulin across individuals²². However, this study did not systematically assess the relative contributions of genetics and the environment²². Understanding how these variables intersect with signal transduction is fundamental to our basic knowledge of signalling and the advancement of personalised medicine, which advocates individualised treatment regimens based on genetic risk factors and gene-by-environment interactions^{16,23}.

Inbred mice allow precise control of genetics and environment unachievable in human studies, enabling examination of how these factors interact to influence biomolecular systems²⁴. Here, we performed phosphoproteomics on insulin-stimulated or control skeletal muscle from five genetically distinct inbred mouse strains fed either an ordinary chow diet or a high-fat, high-sugar “western style” diet. Strikingly, we found that genetic background influenced both the phosphoproteomic insulin response of chow-fed mice, as well as how these responses were modified by high-fat, high-sugar feeding. These signalling changes were reflected in altered activity profiles of multiple kinases and provided insight into the functional organisation of the insulin signalling network by revealing subnetworks of coregulated phosphosites. A major challenge in phosphoproteomics studies is pinpointing important regulatory events among the many responding to a stimulus. We reasoned that associating changes in protein phosphorylation across the gene-by-environment landscape with phenotypic change – in this case insulin-stimulated glucose uptake – would dissect mechanistic targets with greater fidelity. This approach generated known as well as candidate regulators of insulin-stimulated glucose uptake, leading us

to demonstrate that glycolytic upregulation reverses insulin resistance. Our work represents the first global portrait of insulin signalling network plasticity in response to genetic and environmental variation, which will serve as an important resource in future studies of insulin action and resistance.

Results

Phosphoproteomics of insulin signalling in mouse skeletal muscle

To study how protein phosphorylation networks are affected by genetics and environment, we examined insulin signalling in five genetically distinct inbred mouse strains including four lab strains with diverse metabolic phenotypes (C57Bl6J, NOD, BXH9, and BXD34¹⁸), and the wild-derived CAST strain (**Fig. 1a**). Mice underwent a six-week diet regimen of standard lab diet (CHOW) or a high-fat high-sucrose diet (HFD), which is commonly used to induce insulin resistance^{18,25}. Consistent with their diverse genetics, these strains differed in morphometric parameters (body weight, adiposity, lean mass) and metabolic traits (fasting blood glucose, fasting blood insulin, glucose tolerance) both on the CHOW diet and in their response to HFD-feeding (**Fig. S1**).

We focused on skeletal muscle, as it is the site of greatest post-prandial glucose uptake and the most significant contributor to impaired glucose disposal in type 2 diabetes²⁶. Specifically, we chose to examine the soleus muscle, because its largely oxidative fibre composition resembles human muscle tissue more than other murine muscles²⁷. Mice were injected retro-orbitally with saline control or insulin for 10 minutes, and the soleus was collected for phosphoproteomic analysis (**Fig. 1a**). A tritiated 2-deoxyglucose (³H-2DG) tracer was co-injected to measure soleus glucose uptake.

Using the EasyPhos workflow and data-independent acquisition (DIA) mass spectrometry^{28,29}, we quantified 28,809 phosphopeptides across 95 biological samples, corresponding to 23,126 unique high-confidence phosphosites (Class I; localization score > 0.75) on 3,507 proteins (**Fig. 1b**, **Table S1**). On average we quantified 15,395 phosphopeptides in each sample (**Fig. S2a**). Due to the range in soleus mass across strains (**Fig. S1d**) we altered the protein material used for EasyPhos (C57Bl6J and NOD: 755 µg, BXH9 and BXD34: 511 µg, CAST: 364 µg), though phosphopeptide quantification was minimally affected, with only 12.4% fewer phosphopeptides quantified on average in CAST compared to the C57Bl6J/NOD (average 13891.56 vs 15851.29 **Fig. S2a**). Furthermore, while different strains clustered noticeably by the amount of protein material used, samples from animals of the same strain and diet were still highly correlated and generally clustered together, implying the data are reproducible (**Fig. S2b-d**).

To explore the soleus insulin signalling network, we examined phosphopeptides altered by insulin stimulation in at least one of the ten strain-diet combinations. First, to allow comparison across conditions, phosphopeptides were retained if they were quantified well enough to assess the effect of insulin in more than eight strain-diet combinations (≥ 3 insulin-stimulated values and ≥ 3 unstimulated values in each combination). We then tested the resulting 10,432 highly quantified phosphopeptides for significant differences between unstimulated and insulin-stimulated samples (three-way ANOVA insulin main-effect q -value < 0.05) that were of sufficient magnitude in at least one strain-diet combination (insulin/unstimulated fold change > 1.5). This resulted in 441 insulin-regulated phosphopeptides on 232 proteins, which is noticeably more than recent studies of 10-minute insulin signalling in patient-derived myoblasts (174 phosphopeptides³⁰ and 242 phosphopeptides³¹) or mouse adipose tissue (319 phosphopeptides³²) (**Fig. 1b, c**, **Table S1**).

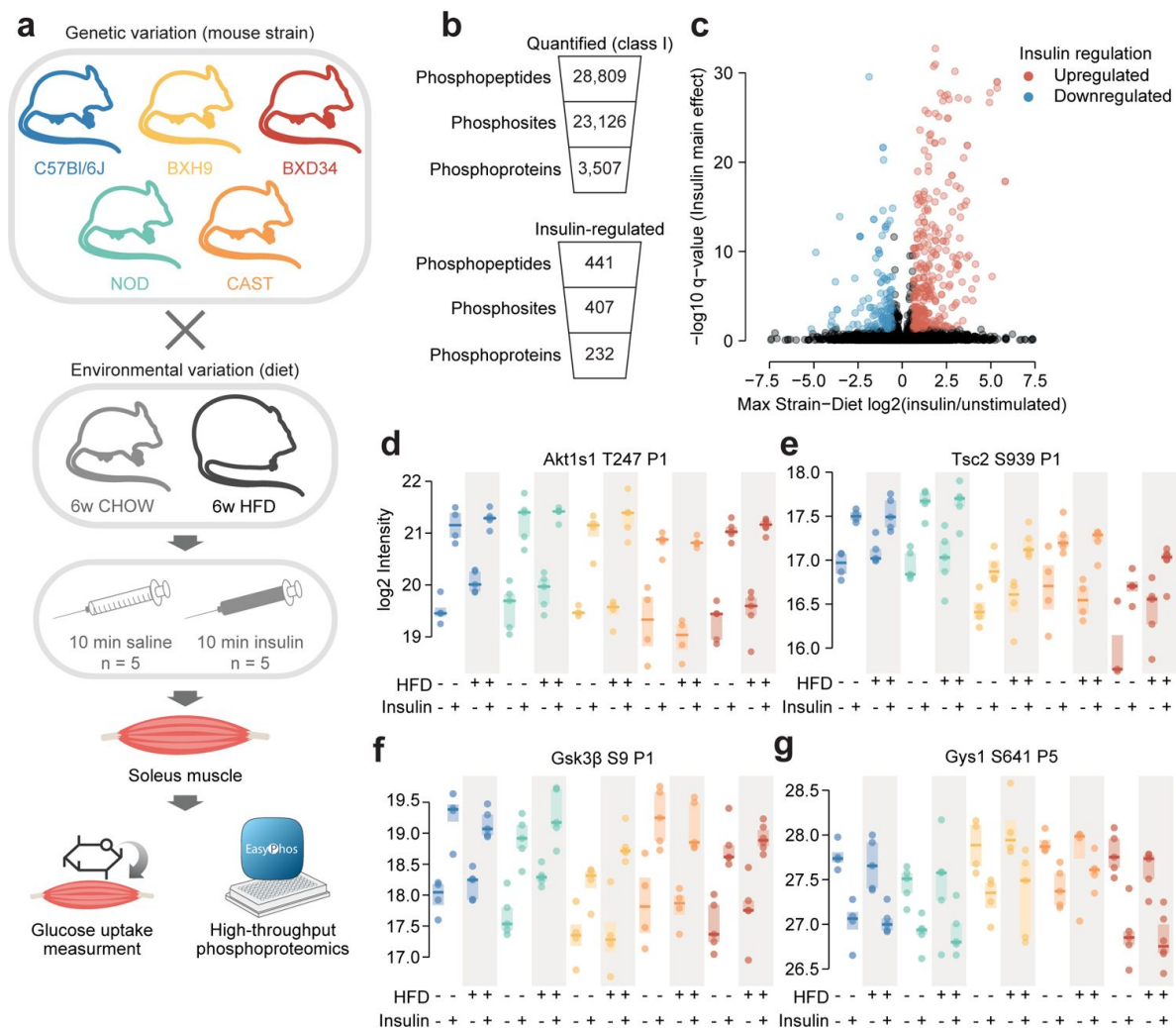


Figure 1

Phosphoproteomics of insulin signalling in mouse skeletal muscle.

a) Workflow for skeletal muscle phosphoproteomics of insulin signalling. **b)** Quantification of skeletal muscle phosphoproteomics. **c)** Volcano plot identifying insulin-regulated phosphopeptides. The greatest $\log_2(\text{insulin/unstimulated})$ fold change across strain-diet combinations is plotted against significance (insulin stimulation main effect, three-way ANOVA). Three phosphopeptides with $-\log_{10} q\text{-values}$ greater than 35 were removed for visual clarity. **d-g)** Example insulin-regulated phosphopeptides. The protein and phosphorylated amino acid are indicated, as well as the number of phosphosites on the phosphopeptide (e.g. "P1"). $n = 4\text{-}6$ biological replicates.

Our analysis recovered many well-studied insulin-regulated phosphosites, including Akt substrates such as T247 on Akt1s1 (PRAS40), S939 on Tsc2, and S9 on Gsk3 β (**Fig. 1d-f**), as well as targets of downstream kinases including the Gsk3 β substrate S641 on Gys1 (**Fig. 1g**). Enrichment of Gene Ontology (GO) biological processes recapitulated canonical insulin signalling axes including “insulin receptor signaling pathway”, “phosphatidylinositol 3-kinase signaling”, “protein kinase B signaling”, and “TORC1 signaling”, and multiple pathways related to glucose metabolism, fatty acid metabolism, autophagy, and protein translation, reflecting known targets of insulin action (**Fig. S3a**). Furthermore, insulin-regulated phosphosites were enriched for insulin regulation in our previous human skeletal muscle phosphoproteome (fold enrichment = 4.22, $p = 9.50 \times 10^{-24}$, one-sided Fisher’s exact test, **Fig. S3b**). Despite this, only half of all insulin-regulated phosphopeptides (228/441) were previously annotated as insulin-regulated in the PhosphositePlus database³³ (**Fig. S3c**), highlighting the potential of our data to discover novel aspects of insulin signalling while recapitulating known components. In addition to insulin, exercise also promotes GLUT4 translocation in skeletal muscle³⁴. We identified a small subset of phosphosites regulated by insulin in this study that were also regulated by exercise in two separate human phosphoproteomics studies^{22,35} (**Fig. S3d**, **Table S2**, phosphosites: Eef2 T57 and T59, Mff S129 and S131, Larp1 S498, Tbc1d4 S324, Svit S300, Gys1 S645), providing a starting point for exploring conserved signalling regulators of GLUT4 translocation. Overall, our phosphoproteomics data provide a comprehensive and high-quality atlas of insulin signalling in mouse skeletal muscle.

Genetics and diet modulate insulin signalling

The influence of genetic and environmental variation on global insulin signalling responses is largely unknown. We therefore developed a pipeline to address this question using our phosphoproteomics data (**Fig. 2a**). First, we converted the intensity values of each insulin-regulated phosphopeptide to “insulin response” values, by normalising insulin-stimulated data to the unstimulated median of the corresponding Strain-Diet combination. Since protein expression should not change within a 10 min insulin stimulation³², this allowed us to parse out protein abundance differences across strains and diets and focus solely on acute signalling processes. We then assessed the impact of genetics in CHOW-fed mice by identifying phosphopeptides with differing insulin responses in one or more strains compared to C57Bl6J (“Strain effect”). Lastly, we explored the effects of HFD-feeding on signalling as two types of “Diet effects”, either as a “Uniform diet effect” – where HFD-feeding affects each strain similarly – or a “StrainxDiet effect” – where its impact depends on the mouse strain. Analyses of “Strain effects” and “Diet effects” were performed separately, so a phosphopeptide could have both a Strain effect and a StrainxDiet/Uniform Diet effect.

Almost half of the 441 insulin-regulated phosphopeptides displayed a Strain effect (**Fig. 2b**, **Table S1**). These included phosphopeptides where C57Bl6J had a stronger insulin response than other strains, such as S15 on the RNA methyltransferase Rnmt (**Fig. 2c**), and phosphopeptides where C57Bl6J had a weaker insulin response, such as S48 on the vesicle fusion regulator Vamp3 (**Fig. 2d**). Vamp3 S48 is predicted to be highly functional (functional score = 0.750³⁶), its phosphorylation correlates with glucose uptake in insulin-stimulated and/or exercised human skeletal muscle²², and Vamp3 overexpression rescues GLUT4 translocation in insulin resistance³⁷, suggesting that this site may represent a genetically variable control point of GLUT4 trafficking. In general, insulin responses were weaker in the four remaining strains compared to C57Bl6J, though the extent of this trend was strain-dependent (**Fig. 2e**). In all, the strain-affected phosphopeptides reveal a unique fingerprint of insulin signalling within each strain (**Fig. 2f**), highlighting the complex and widespread effects of genetic variation on signalling networks.

We next examined the impact of HFD-feeding in insulin signalling. StrainxDiet effects were more prevalent than Uniform diet effects (110 vs 10 phosphopeptides, **Fig. 2g**, **Table S1**), suggesting that the molecular impact of dietary perturbation was strongly modulated by genetic background.

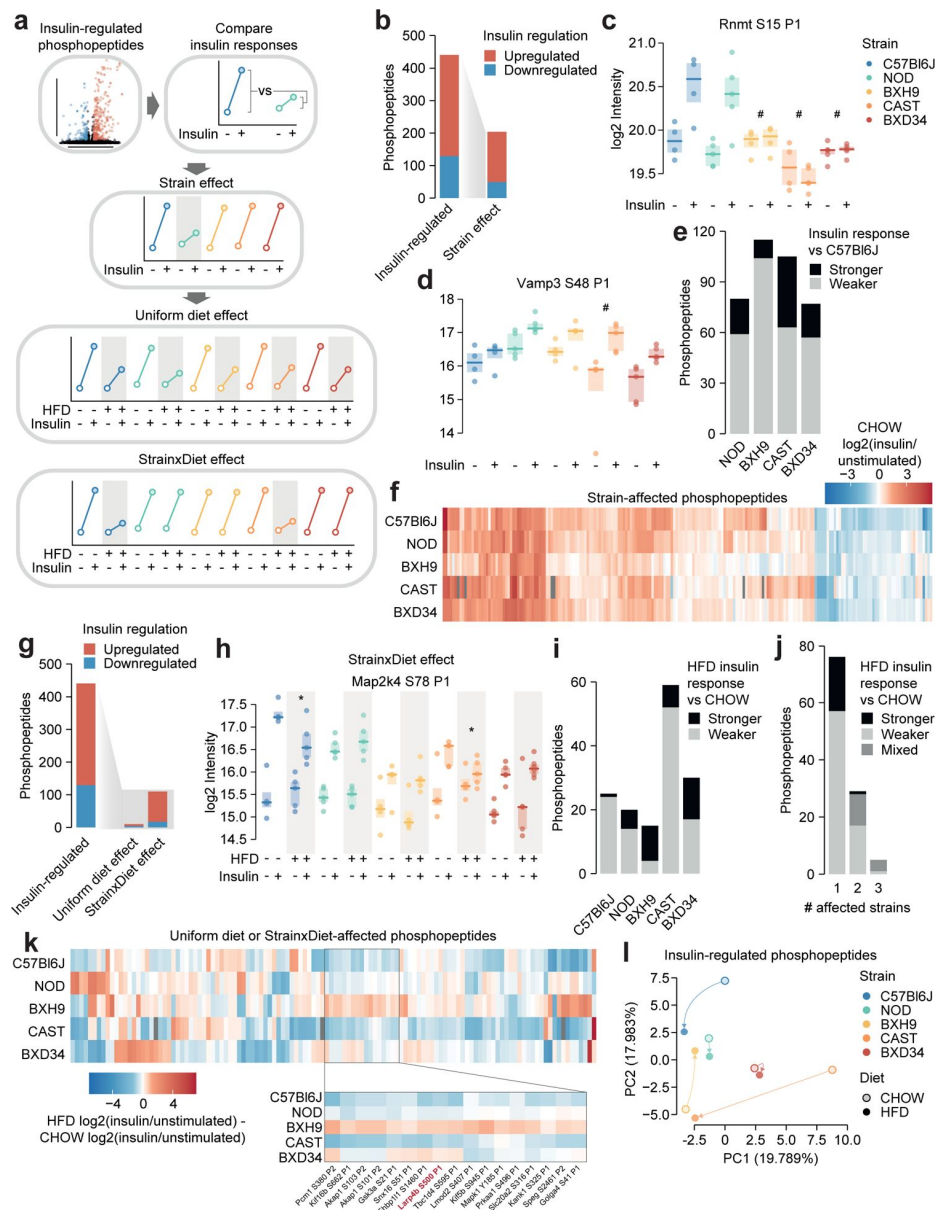


Figure 2

Genetics and diet modulate insulin signalling.

a) Schematic for identifying effects of strain and diet on insulin signalling. **b**) The number of total insulin-regulated phosphopeptides and those with a Strain effect. **c-d**) Two phosphopeptides with Strain effects. ANOVAs were performed on CHOW insulin responses following two-sided t-tests comparing each strain to C57Bl6J (q-values: #). Only CHOW values are shown. **e**) The number of phosphopeptides with stronger or weaker insulin regulation in each strain compared to C57Bl6J. **f**) Heatmap displaying all insulin-regulated phosphopeptides with a Strain effect. Missing values are coloured grey. **g**) The number of total insulin-regulated phosphopeptides and those with diet effects. **h**) A phosphopeptide with a StrainxDiet effect. A two-way ANOVA was performed on insulin response values followed by two-sided t-tests comparing HFD to CHOW within each strain (q-values: *). **i-j**) The number of phosphopeptides with a StrainxDiet effect in **i**) each strain, or **j**) each number of strains. Colour indicates whether the insulin response in HFD is stronger vs CHOW, weaker vs CHOW, or both in different strains ("Mixed"). **k**) Heatmap displaying all insulin-regulated phosphopeptides with a Uniform diet effect or StrainxDiet effect. Inset displays example sites where BXH9 effects contrasted other strains. **l**) PCA of all insulin-regulated phosphopeptides using the log2(insulin/unstimulated) fold changes for each Strain-Diet combination. The percentage of total variance explained by each principal component is indicated. */#: 0.01 ≤ q < 0.05, **/##: 0.001 ≤ q < 0.01, ***/###: q < 0.001. n = 4-6 biological replicates.

StrainxDiet effects impacted known regulatory phosphosites such as the inhibitory site S78 on Map2k4, whose insulin response was attenuated by HFD-feeding only in C57Bl6J and CAST (**Fig. 2h**). Map2k4 activates p38 and JNK kinases which have been implicated as drivers of HFD-induced insulin resistance⁸, and based on the behaviour of S78, the orchestration of this detrimental signalling axis in HFD-feeding may depend on genetic background.

HFD-feeding exerted complex effects on signalling, with the balance between suppressed and enhanced insulin responses varying across strains (**Fig. 2i**). Furthermore, insulin-regulated phosphopeptides were largely altered by HFD-feeding in only a single strain, and when multiple strains were affected, they often changed in opposite directions (**Fig. 2j-k**). For instance, multiple insulin responses were strengthened in BXH9 but weakened in C57Bl6J or CAST (**Fig. 2k**), such as S500 on the translation regulator Larp4b (**Fig. S3e**). Principal component analysis supported the highly divergent impact of dietary perturbation, as HFD-feeding displaced each strain in a distinct direction in principal component space (**Fig. 2l**). StrainxDiet and Strain effects were driven by a mixture of changes to insulin-stimulated phosphorylation, unstimulated phosphorylation, or both, highlighting the complexity of these signalling alterations (**Fig. S3f-g**). This analysis demonstrates the pervasive role of genetics in shaping signalling networks, as genetic background profoundly modulated the effect of HFD-feeding on insulin signalling.

Exploring genetic and dietary modulation of the insulin signalling network

To understand the insulin signalling circuitry and functional pathways modulated by genetics and diet, we curated a network of 160 insulin-regulated phosphosites, comprising sites from a knowledge pathway-derived list of canonical insulin signalling proteins²² and substrates of insulin-regulated kinases (**Fig. 3**, see Methods). Strain and diet affected multiple highly studied signalling sites, including Tsc2 S939 (Diet effect), Gsk3 α S21 (Strain and Diet effects), and Tbc1d4 T649 (Strain and Diet effects), while other sites such as Gsk3 β S9 and Akt1s1 T247 were unaffected. Interestingly, strain and diet affected both canonical and non-canonical insulin signalling proteins to a similar extent (**Fig. S3h-i**). Non-canonical phosphosites could shed light on underappreciated outcomes of insulin action either altered or unaffected by genetics and the environment, such as the p70S6K substrate S47 on Dnajc2 (Strain and Diet effects), which drives cellular senescence³⁸, and S315 on Pcyt1a (no Strain or Diet effect), which inhibits phosphatidylcholine biosynthesis³⁹ (**Fig. 3**).

No functional pathways were overrepresented within strain or diet-affected phosphosites relative to all insulin-regulated sites, implying that genetics and environment modulate most or all outcomes of insulin. For instance, strain and diet affected regulatory phosphosites controlling vesicle trafficking (S348, T575, S595, and T649 on the GLUT4 trafficking regulator Tbc1d4); glucose metabolism (S469 and S486 on Pfkfb2, which promote glycolysis); mitochondrial structure and function (S129 on Mff, which promotes mitochondrial fission); autophagy (S555 on the master autophagy regulator Ulk1); gene transcription (the inhibitory site S108 on the transcription factor Tfeb); and mRNA translation (S236 on ribosomal proteins S6, S422 on Eif4b, and S64 and T69 on Eif4ebp1, which promote translation). Interestingly, Strain and Diet effects overlapped significantly (fold enrichment = 1.50, $p = 4.40 \times 10^{-9}$, two-sided Fisher's exact test, **Fig. S3j**), implying some phosphosites may be more amenable to regulation overall. As a notable exception, all six insulin-regulated phosphosites on Plin1 had Diet effects while only one had a Strain effect (**Fig. 3**). Plin1 coats and regulates lipid droplets, hence this enrichment of Diet effects may represent a signalling response to increased intramuscular lipids in the HFD condition. Overall, genetics and environment triggered widespread alterations in insulin signalling impinging on diverse cellular pathways.

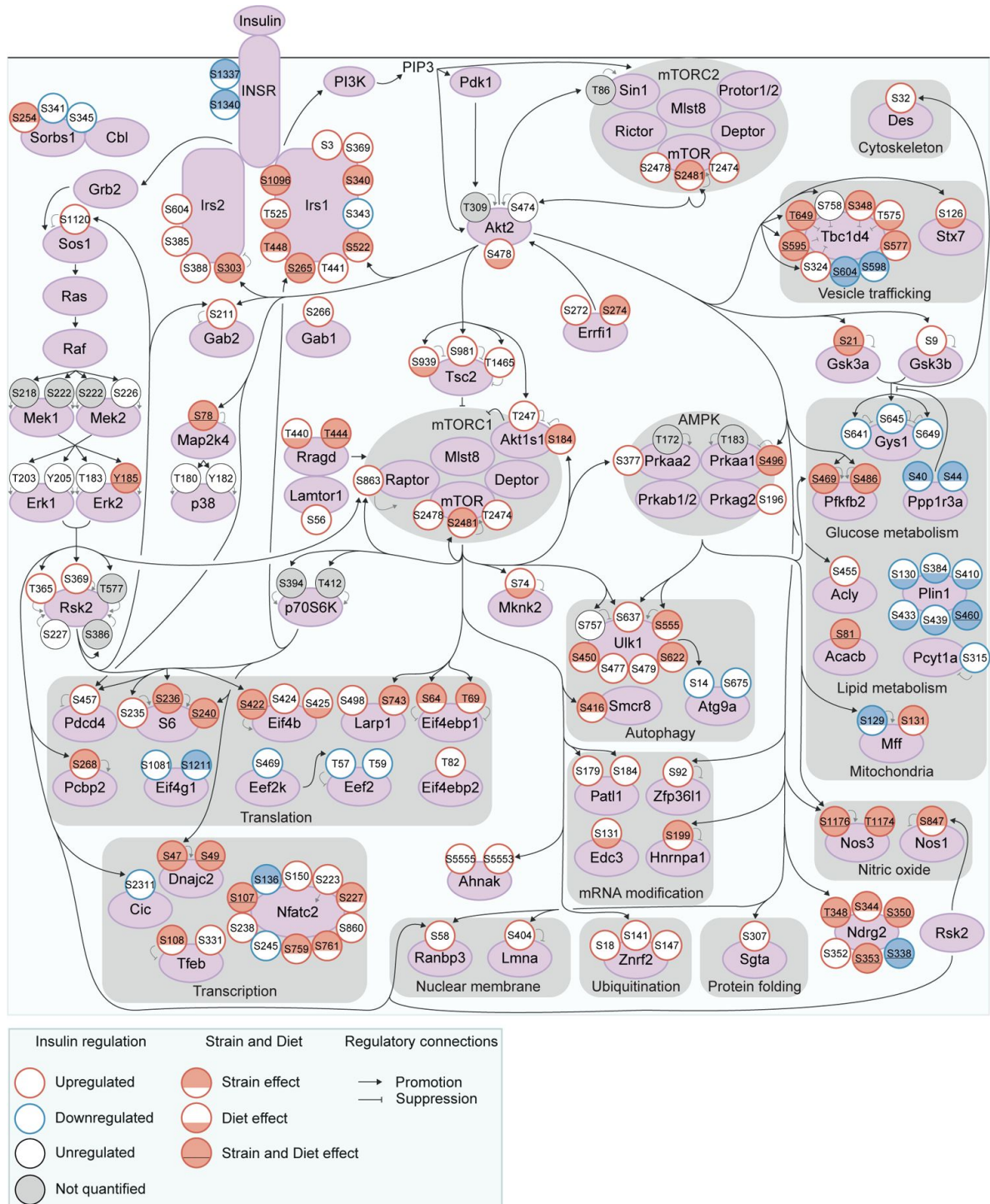


Figure 3

Exploring genetic and dietary modulation of the insulin signalling network.

A curated network of 160 insulin-regulated phosphosites. Phosphosites are depicted as circles where the outline colour denotes the direction of insulin regulation, and the inner colour denotes the presence of Strain effects or Diet effects (either a StrainxDiet or Uniform diet effect). Black arrows indicate regulatory relationships from proteins to other proteins or phosphosites. Grey lines indicate phosphosite regulatory roles.

Genetics and diet rewire insulin-regulated kinase signalling

The extensive signalling changes caused by genetics and diet may result from altered kinase regulation. We tested this hypothesis using a kinase substrate enrichment analysis (KSEA)⁴⁰ on phosphopeptide insulin responses. KSEA accurately captured the activation of canonical insulin-regulated kinases (Akt, mTOR, p70S6K, and p90RSK) and the deactivation of GSK3, confirming the validity of the approach (Fig. 4a). Focussing on CHOW-fed mice, we identified seven kinases differentially enriched across mouse strains (ANOVA adjusted p-value < 0.05, Fig. 4b). For example, insulin activated SGK and deactivated GSK3 more in C57Bl6J and NOD than in other strains (Fig. 4b). Extending this analysis to all mice, we identified kinases with Uniform diet or StrainxDiet effects (Fig. 4c). Akin to our analysis of individual phosphosites (Fig. 2), StrainxDiet effects were more prevalent than Uniform diet effects (five kinases compared to one), indicating that genetic background strongly influences the impact of HFD-feeding on kinase signalling. These results suggest that the observed phosphosite signalling changes could be partly due to altered insulin regulation of multiple kinases.

Biological variation reveals functional organisation of the insulin signalling network

KSEA predicted changes in overall kinase activity, but we questioned if substrates of the same kinase could be differentially regulated by genetic and environmental variation. As a case study we examined substrates of Akt – a master regulator of insulin signal transduction – to assess the similarity of their insulin responses across strains and diets. Strikingly, we observed a range of both positive and negative correlations (Fig. 5a). For instance, while Tsc2 S939 and Akt1s1 T247 both activate mTORC1, their insulin responses correlated poorly ($r = 0.202$, $p = 0.168$, Fig. 5b). Supporting these findings, similar heterogeneity in Akt substrate insulin responses has previously been observed in skeletal muscle from humans with differing metabolic health⁴¹. Hierarchical clustering revealed four distinct groups of positively correlated Akt substrates (Fig. 5a), suggesting these groups may coordinate distinct functional outputs of Akt signalling.

We next explored if the genetic and environmental variation in our study could reveal organisational principles of the entire insulin signalling pathway. By performing weighted gene correlation network analysis (WGCNA)^{42,43}, we identified eight subnetworks of coregulated insulin-responsive phosphopeptides (Fig. 5c) varying in size from 16 to 120 phosphopeptides, with 91 assigned to no subnetwork (Fig. 5d, Table S3). Examining the subnetwork “eigenpeptides” – a weighted average of the constituent phosphopeptides^{42,43} – revealed that the subnetworks captured distinct effects of genetics and diet on insulin signalling (Fig. 5e, Fig. S4a). For example, HFD-feeding attenuated the insulin response of subnetwork I in CAST and C57Bl6J strains (t-test adjusted $p = 0.0256$, 0.0365), while subnetwork II was affected by HFD-feeding only in CAST and NOD (Fig. 5e, Fig. S4a, t-test adjusted $p = 0.00258$, 0.0256). This suggests that the subnetworks may be sensitive to distinct cellular information.

Next, we characterised the regulatory and functional nature of these subnetworks. Canonical insulin-regulated kinases such as Akt and mTOR were enriched across multiple subnetworks (Fig. 5f), confirming that genetic and environmental variation can reveal uncoupling of substrates targeted by the same kinase (Fig. 5a-b). Nevertheless, visualising these subnetworks within our curated insulin signalling pathway (Fig. 3) revealed cases where signal flowed through a single subnetwork, such as from Erk2 (Y185) to its target kinase Rsk2 (T365 and S369) and Rsk2 substrates (Gab2 S211 and Nos1 S847) within subnetwork III (Fig. S5). Within multiply phosphorylated proteins, phosphosites either belonged to the same subnetwork (e.g. Plin1) or diverse subnetworks (e.g. Tbc1d4 and the transcription factor Nfatc2), suggesting the latter may serve as hubs integrating diverse cellular information (Fig. S5).

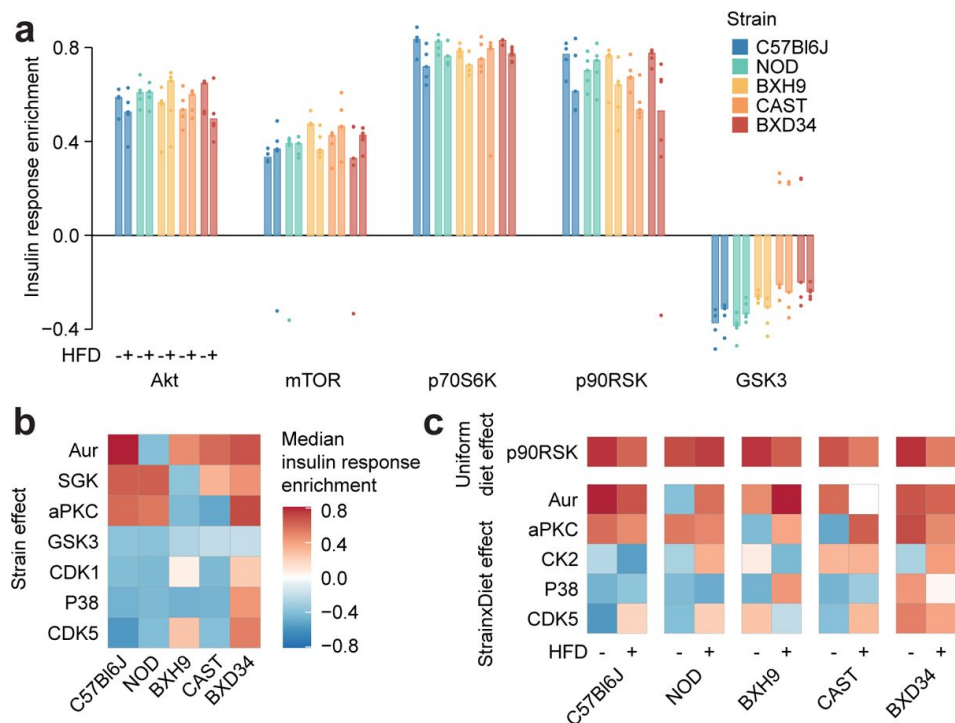


Figure 4

Genetics and diet rewire insulin-regulated kinase signalling.

a) Kinase substrate enrichment analysis (KSEA)³⁷ of five canonical insulin-regulated kinases using insulin response values and kinase-substrate annotations from PhosphositePlus³². **b-c)** Kinase enrichment scores were tested for **b)** Strain effects (CHOW ANOVA adjusted $p < 0.05$) or **c)** StrainxDiet effects (two-way ANOVA interaction effect adjusted $p < 0.05$) and Uniform diet effects (Diet main effect adjusted $p < 0.05$, interaction effect adjusted $p \geq 0.05$). $n = 4-6$ biological replicates.

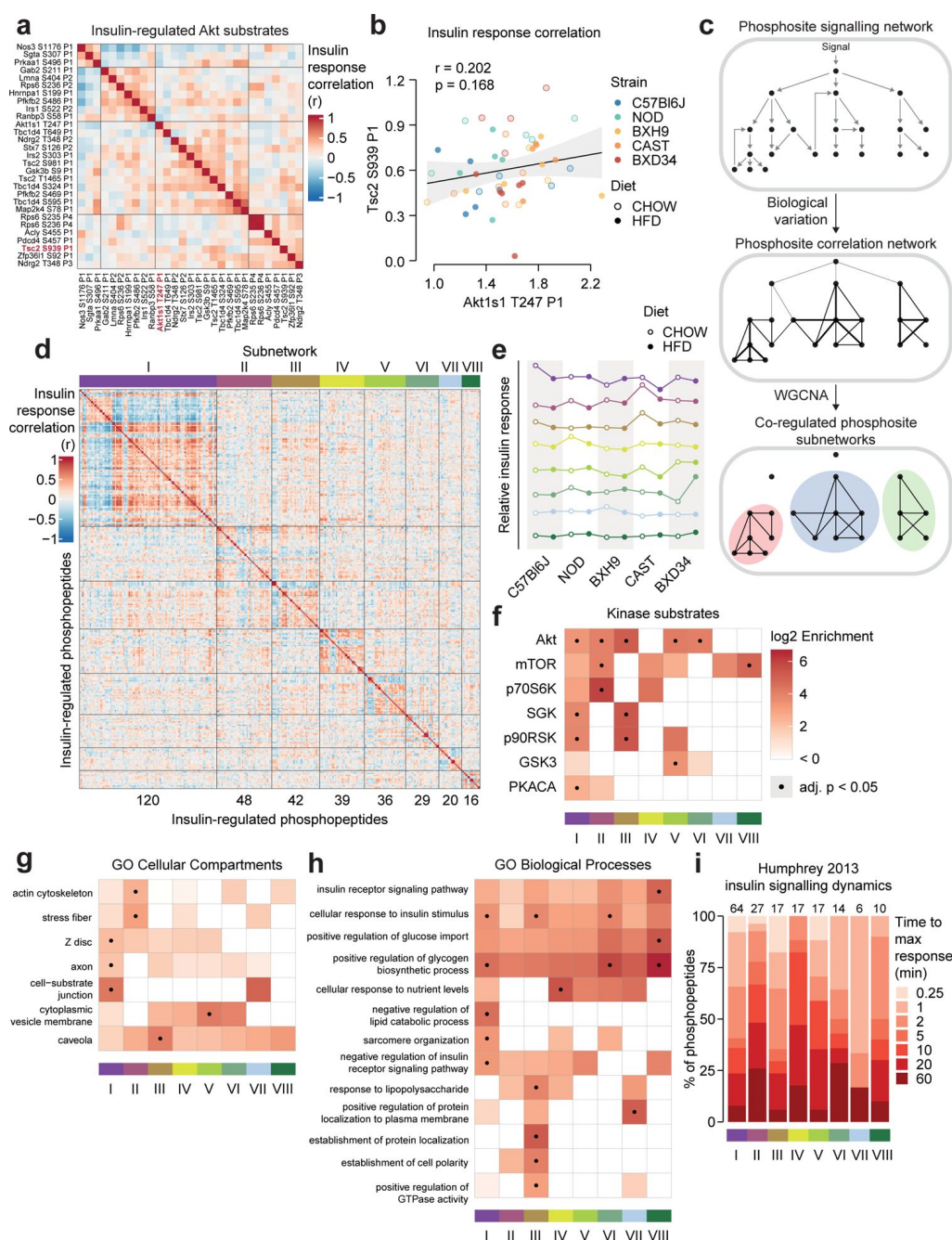


Figure 5

Biological variation reveals functional organisation of the insulin signalling network.

a) Pairwise Pearson's correlation of the insulin response values of insulin-regulated Akt substrates. Substrates were separated into four clusters by hierarchical clustering followed by tree cutting. **b)** The correlation between insulin response values of the Akt substrates Tsc2 S939 and Akt1s1 T247. Linear regression is indicated with 95% confidence intervals. **c)** Rationale for performing WGCNA. **d)** Pairwise Pearson's correlation of all insulin-regulated phosphopeptides separated into WGCNA-derived subnetworks. The number of phosphopeptides in each subnetwork is indicated below the heatmap. **e)** The "eigenpeptide" of each subnetwork. The median of each strain-diet combination is shown. **f-h)** The enrichment of **f)** PhosphoPlus-derived kinase-substrate annotations³², **g)** GO cellular compartments, and **h)** GO biological processes within each subnetwork relative to the entire phosphoproteome (one-sided Fisher's exact test, Benjamini-Hochberg p-value adjustment). **i)** The time taken for phosphopeptides to reach maximum insulin-stimulated intensity in a previous study of insulin signalling dynamics¹⁰. The number of phosphopeptides mapped into the study is indicated above each bar.

Some subnetworks were enriched in specific cellular compartments (GO ontologies), implying that common localisation may facilitate coregulation of phosphosites (**Fig. 5g**). Coregulation may partition functional outcomes of insulin action, as certain biological processes were enriched only in select subnetworks (**Fig. 5h**). These included known insulin targets like “negative regulation of lipid catabolic process” in subnetwork I and “positive regulation of glycogen biosynthetic process” in I, VI, and VIII (**Fig. 5h**). To further probe functional differences we analysed phosphopeptide subnetwork membership scores, which revealed additional pathways enriched in individual subnetworks. However, these results were not significant after p-value adjustment and hence are suggestive only (**Fig. S4b-d**). Lastly, we leveraged our previous phosphoproteomic time course of insulin signalling to interrogate subnetwork dynamics¹⁰ and found that phosphopeptide insulin response dynamics varied across subnetworks (**Fig. 5i**). This reveals distinct temporal regulation as another feature underlying the substructure of the insulin signalling network. Overall, genetic and environmental diversity illuminated the complex coregulation structure of insulin signalling, featuring subnetworks that evade known network circuitry and present unique functional signatures.

Leveraging biological variation to identify drivers of insulin responsiveness

We have so far described the marked influence of genetic background and HFD-feeding on skeletal muscle insulin signalling, evident at the level of individual phosphosites, protein kinases, and co-regulated network modules. We hypothesised that by associating this signalling diversity with a phenotypic output of insulin, such as enhanced glucose uptake, we would filter out mechanistically irrelevant phosphosites and hence prioritise molecular regulators of the phenotype. To test this hypothesis, we measured *in vivo* glucose uptake with ³H-2DG tracer in the same muscle samples used for phosphoproteomics. Insulin-stimulated glucose uptake differed by more than twofold across strains (two-way ANOVA strain effect $p = 4.78 \times 10^{-7}$) and was almost uniformly decreased by HFD-feeding (two-way ANOVA diet effect $p = 1.83 \times 10^{-5}$) (**Fig. 6a**). This highlights that genetic background and dietary status are key determinants of insulin responsiveness.

To prioritise signalling nodes responsible for differences in insulin responsiveness, we correlated all insulin-regulated phosphopeptides with glucose uptake in insulin-stimulated muscles, resulting in 37 significantly correlated phosphopeptides ($r > 0.35$ or < -0.35 , $q\text{-value} < 0.1$, **Fig. 6b**). The most significantly correlated phosphopeptide contained T1174 and S1176 on the nitric oxide synthase Nos3. The latter serves as a positive control for our analysis, as this site is known to be phosphorylated in response to insulin to promote *in vivo* glucose uptake by vasodilation^{44–46} (**Fig. 6c**). Other correlated phosphopeptides that could modulate insulin responsiveness include S1751 on Afdn, a phosphosite implicated in insulin action⁴⁷, and S196 on the Prkag2 subunit of AMPK, a major metabolic signalling hub promoting glucose uptake⁴⁸ (**Fig. 6b**). These examples suggest that our analysis prioritised regulators of glucose uptake.

While the above analysis identified phosphosites associated with glucose uptake through their absolute abundance, we hypothesized that for some phosphosites, the magnitude of their response to insulin may be a stronger determinant of insulin action. We found that the insulin response values of 13 phosphopeptides correlated with insulin-stimulated glucose uptake ($r > 0.35$ or < -0.35 , $q\text{-value} < 0.1$, **Fig. 6d**). These were largely distinct from the 37 phosphopeptides identified in our first analysis, indicating that the two approaches captured complementary information. Several of these phosphopeptides could regulate insulin-stimulated glucose uptake, such as the regulatory site S469 on the enzyme Pfkfb2 which activates glycolysis, a major pathway for glucose consumption^{49–51} (**Fig. 6e**), and S177 on Rcsd1, which could affect GLUT4 vesicle transport via actin cytoskeleton remodelling⁵² (**Fig. 6d**). The associations identified in these analyses could arise uniquely at the level of protein phosphorylation or could be driven by changes in total protein abundance. To assess this, we mined soleus total proteomics data from a recent study of

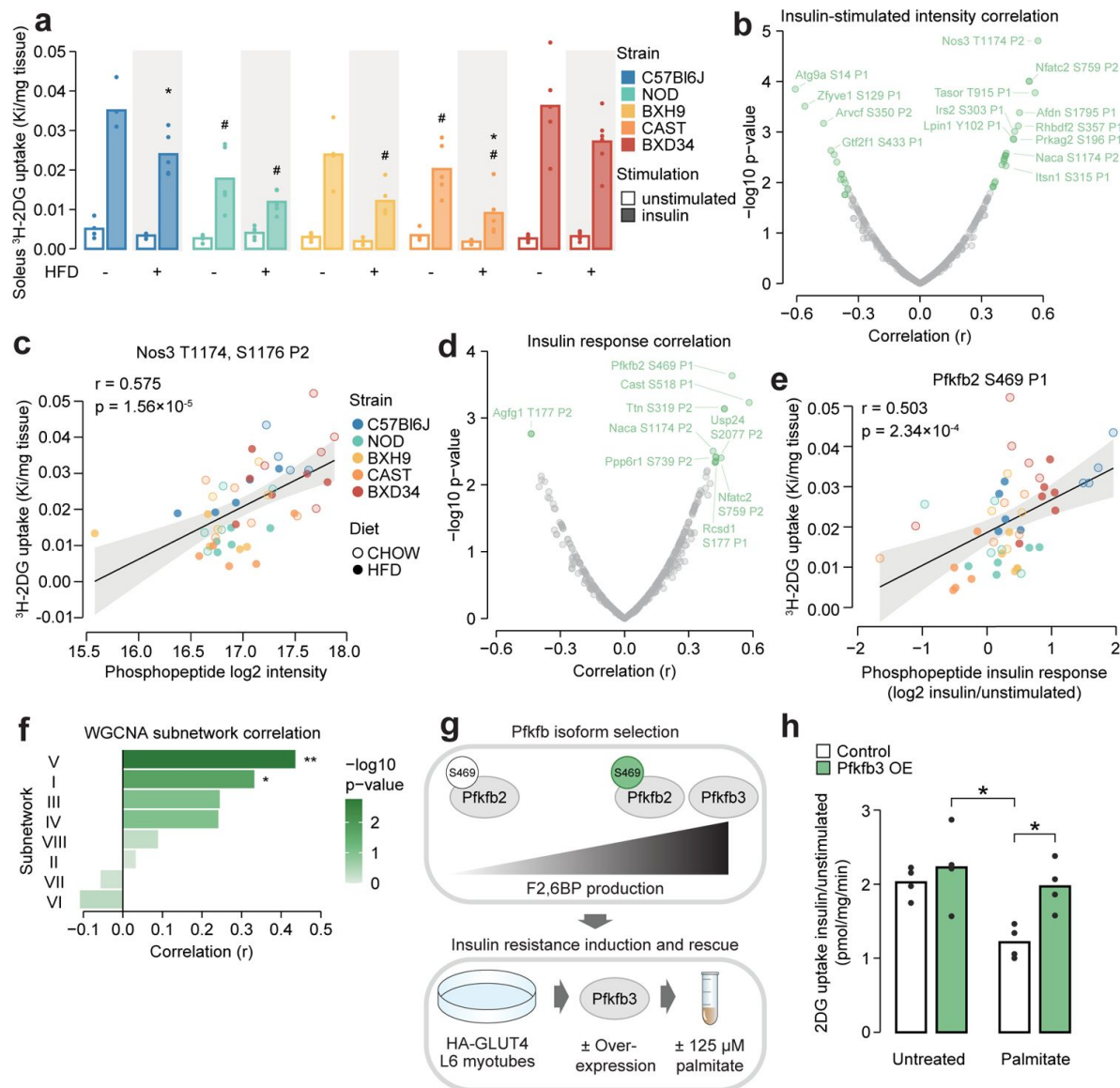


Figure 6

Leveraging biological variation to identify drivers of insulin responsiveness.

a) The uptake of ^3H -2DG into mouse soleus muscle after a 10 min injection of insulin (1 U/kg lean mass; "insulin") or saline ("unstimulated") calculated as a rate constant (Ki). Two-sided t-tests were performed on insulin-stimulated uptake values to compare HFD to CHOW within each strain (adjusted p-value: *) or each strain to C57Bl6J within either diet (adjusted p-value: #). $n = 4$ -6 biological replicates. **b)** Pearson's correlation between log2 intensity of insulin-regulated phosphopeptides and ^3H -2DG uptake within insulin-stimulated mice. Significantly correlated phosphopeptides (q-value < 0.1, $r > 0.35$ or $r < -0.35$) are coloured green and select correlated phosphopeptides are labelled. **c)** Correlation of Nos3 T1174, S1176 insulin-stimulated intensity with insulin-stimulated ^3H -2DG uptake. Linear regression is indicated with 95% confidence intervals. **d)** As in **b)**, using phosphopeptide insulin response values. **e)** Correlation of the Pfkfb2 S469 insulin response with insulin-stimulated ^3H -2DG uptake. **f)** Correlation of WGCNA subnetwork eigenpeptides with insulin-stimulated ^3H -2DG uptake. Significant correlations are indicated (*). **g)** Rationale and workflow for over-expressing Pfkfb3 to rescue palmitate-induced insulin resistance. **h)** The fold change of unstimulated to insulin-stimulated glucose uptake (100 nM insulin, 20 min) in L6-GLUT4-HA myotubes with or without Pfkfb3 overexpression, treated with palmitate (125 μM , 16 h) or BSA vehicle control. A two-way ANOVA was performed followed by Tukey's posthoc tests (*). Not all significant comparisons are shown. $n = 4$ biological replicates. */#: 0.01 $\leq p < 0.05$, **/##: 0.001 $\leq p < 0.01$, ***/###: $p < 0.001$

seven CHOW and HFD-fed mouse strains, three of which were in common with this study (C57BL6J, BXH9, BXD34)¹⁸. Of the 34 glucose uptake-associated phosphoproteins we identified, 16 were quantified in both studies and only two additionally correlate with glucose uptake at the total protein level (Ppp6r1 and Ttn, **Fig. S6a**). Hence, it seems likely that most glucose uptake-phosphosite associations were not driven by protein abundance changes, underscoring the utility of phosphoproteomics to provide unique information on top of more classical omics layers.

In addition to individual phosphosites, the status of larger signalling network components could also influence insulin responsiveness. Kinase enrichment scores affected by strain or diet did not correlate with glucose uptake (**Table S4**), suggesting insulin action is not dominated by the net activity of specific kinases. Interestingly, two WGCNA-derived insulin signalling subnetworks correlated with glucose uptake: subnetwork V ($r = 0.436$, $p = 0.00173$) and subnetwork I ($r = 0.332$, $p = 0.0197$, **Fig. 6f**). Subnetwork V could modulate glucose uptake through actin cytoskeleton remodelling via Rscd1 S177, through glucose metabolism promotion via Gys1 S641 (**Table S3**), and by influencing GLUT4 vesicle trafficking due to its enrichment at “cytoplasmic vesicle membranes” (**Fig. 5g**). It was also enriched in substrates of GSK3, which has been implicated in insulin resistance in skeletal muscle^{53–55} and adipose tissue³². Subnetwork I, the largest cluster containing 27% of insulin-regulated phosphopeptides, was enriched in multiple kinases and biological processes (**Fig. 5f, h**), suggesting it may be a central regulatory hub for various outcomes of insulin action including glucose uptake. Examining the subnetwork membership scores for glucose-uptake correlated phosphopeptides also revealed a preference for clusters V and I, supporting this analysis (**Fig. S6b–c**). Overall, compartmentalisation of insulin-responsive phosphosites into subnetworks may enable independent control of insulin’s functional outputs, since only two subnetworks correlated with insulin-stimulated glucose uptake.

Upregulating glycolysis reverses insulin resistance

We next aimed to validate our approach for identifying regulatory mechanisms of insulin-stimulated glucose uptake. S469 on Pfkfb2 correlated highly with glucose uptake follow insulin stimulation (**Fig. 6e**). Phosphorylation of this site leads to increased production of F2,6BP, a potent glycolytic agonist, suggesting that activating glycolysis may play a key role in muscle insulin responsiveness. This is consistent with our previous findings that glycolytic enzyme abundance was strongly associated with *ex vivo* insulin-stimulated glucose uptake in muscle from inbred mice¹⁸, that the insulin resistance-reversing small molecule thioestrepton enhances glycolytic capacity⁵⁶, and that decreasing glycolytic flux caused insulin resistance *in vitro*⁵⁷. To further establish glycolysis as a regulator of insulin responsiveness in skeletal muscle, we decided to investigate whether upregulating glycolysis through F2,6BP production can restore insulin-stimulated glucose uptake in insulin resistance.

Since Pfkfb2 requires phosphorylation by Akt to produce F2,6BP substantially, increasing F2,6BP production via Pfkfb2 would require enhanced activating site phosphorylation, which is difficult to achieve in a targeted fashion, or phosphomimetic mutation of activating sites to aspartate/glutamate, which often does not recapitulate the molecular effects of serine/threonine phosphorylation⁵⁸. By contrast, the paralog Pfkfb3 has high basal production rates and lacks an Akt motif at the corresponding phosphosites⁵¹. We therefore rationalised that overexpressing Pfkfb3 would robustly increase F2,6BP production and enhance glycolysis regardless of insulin stimulation and Akt signalling (**Fig. 6g**). To avoid systemic effects of Pfkfb3 overexpression we studied cultured L6-GLUT4-HA myotubes, which display robust insulin regulation of GLUT4 trafficking and develop insulin resistance upon palmitate treatment, mimicking lipotoxicity, a trigger of *in vivo* insulin resistance⁵⁹. While L6 cells are of rat origin, they are preferable to the popular C2C12 mouse cell line since the latter lack an insulin-responsive vesicular compartment⁶⁰ and undergo spontaneous contraction, resulting in confounding non-insulin dependent glucose uptake⁶¹.

As anticipated, Pfkfb3 overexpression increased glycolytic capacity in L6-GLUT4-HA myotubes as measured by extracellular acidification rate (**Fig. S7a-c**). Pfkfb3 overexpression also restored insulin-responsive glucose uptake to normal levels in palmitate-treated cells (**Fig. 6h**, **Fig. S7d**). This effect was only observed in cells treated with palmitate and insulin, suggesting it specifically modulated insulin action rather than non-specifically increasing glucose uptake through enhanced glucose consumption (**Fig. S7d**). This may be due to enhanced insulin signalling, as we have previously observed that transient Pfkfb3 overexpression increased Akt signalling in HEK293 cells⁵⁷. However, immunoblotting of canonical insulin-responsive phosphosites on Akt and its substrates GSK3 α/β and PRAS40 revealed minimal effect of palmitate treatment and Pfkfb3 overexpression (**Fig. S7e-f**), hence more detailed phosphoproteomics studies are needed to clarify whether Pfkfb3 overexpression restored insulin action by modulating insulin signalling. Overall, our results further establish glycolytic flux as a major determinant of the glucose uptake arm of muscle insulin action and highlight the power of studying phosphoproteomics across the gene-by-environment landscape to identify causal drivers of complex phenotypes. We anticipate that our catalogue of glucose uptake-correlated phosphosites will provide a rich starting point for future investigations into mechanisms of insulin action and resistance.

Discussion

The environment shapes the flow of information from genotype to phenotype. Many studies have interrogated the role of intermediate molecular layers such as the transcriptome or proteome, however few studies have examined how protein post-translation modifications participate in this information transfer. Here we have approached this problem by leveraging diverse inbred mouse strains and phosphoproteomics to examine the insulin signalling network across a landscape of genetic and dietary variation. Genetic background significantly altered the insulin signalling network both independently and in concert with dietary status, affecting myriad phosphosites and multiple kinases. We exploited this variation in signalling responses in two ways – to study the partitioning of the Akt and insulin signalling pathways into distinct subnetworks of coregulated phosphosites; and to identify potential regulators of insulin responsiveness by associating phosphorylation with insulin-stimulated glucose uptake. Finally, validation studies in L6 myotubes confirmed the major role of accelerated glycolysis as a key regulator of insulin responsiveness.

Genetic and diet-driven signalling changes did not transmit linearly through our current model of the insulin signalling network, illustrating that this model remains incomplete. Notably, substrates of kinases such as Akt clustered into distinct groups based on differing insulin responses. Hence, it is an oversimplification to model signalling pathways as networks of individual kinases since substrates of the same kinase display independent regulation. This could arise from localisation of a kinase to distinct substrate pools^{62–64}; interactors targeting a kinase to different substrates⁶²; substrate phosphorylation by alternate kinases⁶⁵; the dephosphorylation of specific substrates by phosphatases; kinase post-translational modifications altering substrate specificity; and distinct substrate phosphorylation kinetics¹⁰. As our knowledge of the repertoire of kinase substrates continues to deepen⁶⁶, future research should explore how the above mechanisms contribute to finer regulation of these substrates. Genetic and environmental variation also exposed a coregulation subnetwork structure within the insulin signalling network. The enrichment of subnetworks in distinct biological processes, and the selective association of two subnetworks with glucose uptake, suggests that this coregulation structure may direct independent control of distinct outcomes of insulin action. This exciting possibility necessitates further investigation, including replication in independent cohorts and spatiotemporal characterisation of subnetwork dynamics. Alternatively, one could overlap subnetworks with genetic information, such as genes associated with glucose homeostasis and other metabolic traits

in human GWAS studies⁶⁷, or muscle-specific eQTLs or pQTLs genetically colocalised with similar traits⁶⁸, to further prioritise subnetwork-associated phenotypes and identify potential drivers within subnetworks.

Muscle insulin signalling responses vary across individuals^{22,41}, and our results suggest that baseline genetic differences and an individual's environment both alter signalling, with the environment's influence depending strongly on genetic background. Signalling pathways are popular therapeutic targets due to their importance in human health and the relative ease of pharmaceutical interventions⁶⁹. Our results advocate for a personalised approach to such therapies, implying that the efficacy of treatments aiming to correct pathological signalling responses will depend on an individual's genetic background. In cancer, where signalling networks are dysregulated heterogeneously, modelling patient-specific networks has already shown promise for identifying personalised drug targets^{70,71}. Personalised medicine approaches will also be aided by a comprehensive understanding of how genetics shape signalling networks and potentiate their modulation by the environment. Recent studies have made the first step, revealing that the ground-state phosphoproteome can be altered by mutations affecting network components such as kinases, phosphatases, and phosphoproteins, as well as the molecular milieu the network is exposed to including extracellular signalling ligands^{20,21}. An important corollary of such genetic factors is that multiple genetic backgrounds should be studied when establishing generalizable signalling responses. Our data indicate that insulin responses in C57BL6J – the most commonly studied mouse strain – are not necessarily generalizable, indicated by phosphosites such as S15 on Rnmt that were insulin regulated almost exclusively in C57BL6J.

A major challenge in studying signal transduction with omics technologies is that hundreds to thousands of molecular events typically respond to a cellular signal, making it difficult to pinpoint the most crucial regulatory nodes. To tackle this challenge, we previously demonstrated that associating phosphoproteomics with physiological phenotypes across diverse individuals enriches for phosphosites more likely to modulate biological responses²². Here we have elaborated on this approach, revealing that associating phosphorylation with phenotype across a genetic and environmental landscape can identify regulators of specific biological processes, such as insulin-stimulated glucose uptake. Our results recapitulated known glucose uptake regulators and led to further validation of glycolytic flux as a modulator of insulin responsiveness. We have previously demonstrated that reduced glycolytic flux impairs GLUT4 translocation and insulin signalling⁵⁷, implying that the status of glycolysis is sensed by proteins regulating insulin action. An enhanced glycolytic metabolic tone may alter production of reactive oxygen species, a known modulator of insulin action^{59,72,73} and insulin signalling⁷⁴. Alternatively, recent approaches to map protein-metabolite interactions could identify points of allosteric crosstalk between glycolytic metabolites and insulin action proteins^{75,76}, hence broadening our understanding of the bidirectional communication between insulin action and metabolism.

It was striking that only several dozen insulin-regulated phosphopeptides associated significantly with glucose uptake. Since insulin triggers multiple distinct cellular outcomes, it is possible that only a subset of insulin-responsive phosphosites contribute to enhanced glucose uptake. Moreover, many of these phosphosites might only be permissive for insulin-stimulated glucose uptake and are not major regulatory nodes determining the fidelity of the process. For example, while mutation of the four primary Akt regulatory sites on Tbc1d4 blocks GLUT4 translocation⁷⁷, none of these phosphosites featured strong positive correlations with glucose uptake in our analysis (**Fig. S8**). This implies that their phosphorylation may promote glucose uptake in a functionally permissive, switch-like fashion. We suggest that the glucose uptake-associated phosphosites we have identified will be enriched in major regulators of insulin responsiveness, necessitating future functional studies to characterise these sites and explore their involvement in insulin resistance.

Our work demonstrates that genetic and environmental variation can profoundly modulate global signalling networks and that the influences of these factors are intrinsically entwined. We show that the resulting diversity in signalling responses can be leveraged to pinpoint regulators of insulin-stimulated glucose uptake, providing a powerful methodological framework for interrogating the regulatory basis of complex biological pathways.

Limitations of this study

First, this study focused on male mice and examined only five inbred strains. This limited number of strains may mean that our association analysis was underpowered to detect some regulators of insulin responsiveness. Importantly, however, this does not imply that the regulators identified are incorrect, but only that there may be more to discover with larger cohorts. In addition, studies incorporating male and female subjects found that mouse insulin sensitivity was affected by interactions between sex and strain⁷⁸, and insulin signalling in stem cell-derived myoblasts by interactions between sex and donor metabolic health³¹. Hence, we anticipate that sex would contribute another important dimension to the web of interactions between genetics, diet, and insulin signalling identified here. Future work should therefore extend our approach across a broader range of genetic backgrounds, as well as in female mice. Second, we only examined insulin signalling after 10 minutes, since measuring multiple timepoints would have drastically increased the number of animals and samples required. Integration of dynamic phosphoproteome data from cultured cells indicated that insulin signalling dynamics may contribute to trends in our data (**Fig. 5i**), suggesting the exploration of signalling at additional timepoints may be fruitful in the future.

Third, mammalian tissues are a heterogeneous mixture of cell types, and differences in this mixture could result in different signalling responses measured at the whole tissue level. In our experience, the soleus can be reproducibly dissected as an intact muscle with little contamination from surrounding tissues, making it unlikely that cell type composition varied across samples due to tissue collection. However, we cannot exclude the possibility that differences in the composition of the soleus muscle across strains and diets contributed to the signalling changes we detected. We were also unable to determine the extent to which signalling changes arose from muscle-intrinsic or extrinsic factors. For instance, body weight varied substantially across mice and correlated significantly with 25% of Strain and Diet-affected phosphopeptides (**Fig. S8c**), suggesting obesity-related systemic factors likely impact a subset of the muscle insulin signalling network. Furthermore, genetic differences in lifespan could alter the “biological age” of different strains and their phosphoproteomes, though we could not assess this possibility since lifespan data is not available for most strains used. Lastly, as we did not perform total proteomics, we did not assess whether phosphosite changes were caused by differences in total protein abundance. However, since protein abundance should not change within a 10-minute insulin stimulation³², the acute insulin responses we observed should only reflect differences in protein phosphorylation. Furthermore, in our previous studies of insulin signalling in adipocytes³² or human skeletal muscle²² in which deep proteomes were measured in parallel, we found little global correlation between changes in protein phosphorylation and protein abundance across experimental conditions regardless of insulin stimulation, suggesting the contribution of protein abundance to phosphosite changes across strains and diets was likely minimal.

Additional information

Acknowledgements

D.E.J. is an Australian Research Council (ARC) Laureate Fellow. The content is solely the responsibility of the authors and does not necessarily represent the official views of the ARC. The authors also acknowledge the facilities, and the scientific and technical assistance of the Sydney Mass Spectrometry Facility and the Laboratory Animal Services at the Charles Perkins Centre, University of Sydney.

Author Contributions

Conceptualization: JvG, SWCM, SJH, DEJ. Methodology: JvG, HBC, SWCM, ADV, MP, SJH. Formal analysis: JvG. Investigation: JvG, SWCM, HBC, ADV, MP, JS, SM, MEN, SJH. Resources: JS. Writing – Original Draft: JvG. Writing – Review and Editing: All authors. Visualization: JvG. Supervision: SWCM, SJH, DEJ. Project Administration: JvG, SWCM, SJH, DEJ. Funding Acquisition: DEJ

Declaration of Interests

The authors declare no competing interests.

Data Access Statement

All raw and Spectronaut processed phosphoproteomics data have been deposited in the PRIDE proteomeXchange repository and will be made publicly available upon publication. Processed data are available as supplementary tables.

Code availability

All code used to analyse data and produce figures has been uploaded to https://github.com/JulianvanGerwen/GxE_muscle_phos 

Methods

Statistical analysis

Most statistical analysis was performed in the R programming environment using RStudio (R version: 4.2.1, RStudio version: 2022.07.1 Build 554). Analysis of GLUT4-HA-L6 myotube Pfkfb3 expression, 2DG uptake, and ECAR was performed in GraphPad Prism (version: 9.3.1).

Animal details

Male C57BL/6J (C57BL6J), BXH9/TyJ (BXH9), BXD34/TyJ (BXD34), and CAST/EiJ (CAST) mice were purchased from Australian BioResources (Moss Vale, NSW, Australia) while NOD/ShiLtJ (NOD) mice were purchased from Animal Resources Centre (Murdoch, WA, Australia). Mice were at most 9 weeks old upon arrival. Mice were housed at 23 °C on a 12 h light/dark cycle in cages of 2-5, with free access to food and water. At 12-16 weeks of age mice were randomly allocated to a standard CHOW diet (13% calories from fat, 65% from carbohydrate, 22% from protein; “Irradiated Rat and Mouse Diet”, Specialty Feeds, Glen Forrest, WA, Australia) or a high-fat high-sucrose diet made in house (HFD; 45% calories from fat (40% calories from lard), 35% from carbohydrate (14% calories from starch), and 22% from protein) and sacrificed exactly 6 weeks later. The number of mice in

each group are C57Bl6J: 8 CHOW, 10 HFD; NOD: 10 CHOW, 10 HFD; BXH9: 8 CHOW, 9 HFD; CAST: 9 CHOW, 9 HFD; BXD34: 10 CHOW, 11 HFD. Procedures were carried out with approval from the University of Sydney Animal Ethics Committee following guidelines issued by NHMRC (Australia).

Assessment of body composition

Body composition of individual mice was assessed using the EchoMRI-900 to determine lean mass 1 day before a glucose tolerance test and 5-6 days before euthanasia. Analysis was performed as per the manufacturer's specifications.

Glucose tolerance test

On the day of a glucose tolerance test mice were fasted for 6 h (0800-1400). Mice were then orally gavaged with 20% (w/v) glucose in water at 2 g/kg lean mass, and blood glucose was measured from the tail vein using a glucometer 0, 15, 30, 45, 60, and 90 min after the gavage. At 0 and 15 min, 5 μ L blood was also collected into an Insulin Mouse Ultra-Sensitive ELISA plate (Crystal Chem USA, Elk Grove Village, Illinois, USA). Blood insulin concentration was measured according to the manufacturer's protocol, using linear extrapolation from an insulin standard curve. The area of the blood glucose curve (AOC) was calculated by:

$$AOC = \sum_{i=2}^n \frac{(G_{i-1} - G_1) + (G_i - G_1)}{2} (t_i - t_{i-1})$$

Where i represents the i^{th} timepoint at which blood glucose was measured, n represents the last timepoint, t_i represents the time (min) of the i^{th} timepoint, and G_i represents blood glucose concentration (mM) at the i^{th} timepoint.

In vivo insulin stimulation

On the day of the procedure mice were fasted for 2h (1100-1300). Mice were then anaesthetised by intraperitoneal injection of sodium pentobarbital at 80 mg/kg body mass. To counter anaesthesia-associated declines in body temperature, mice were wrapped in aluminium foil and placed on a heating pad at 37°C. After 15 min anaesthesia, mice were injected retro-orbitally as previously described⁷⁹ with 50 μ L plasma replacement (B. Braun, Melsungen, DEU) containing 10 μ Ci ^3H -2DG and saline or insulin (0.75 U/kg lean mass). Blood glucose was measured from the tail vein using a glucometer (AccuCheck, Roche Diabetes Care, NSW, Australia) 1 min prior to injection and 1, 5, 7.5, and 10 min after injection. Simultaneously, 5 μ L blood was collected into 95 μ L 0.9% NaCl on ice to measure ^3H -2DG blood concentration. Ten minutes after insulin injection mice were sacrificed by cervical dislocation and the soleus muscle was rapidly excised, immediately frozen in liquid nitrogen, and stored at -80°C. To measure ^3H -2DG blood concentration, diluted blood samples were first centrifuged at 10,000 xg for 10 min to pellet blood cells. Supernatant (70 μ L) was collected and combined with 3 mL liquid scintillation cocktail (Perkin Elmer, Massachusetts, USA: 6013321) to allow the measurement of ^3H with a Tri-Carb 2810TR Liquid Scintillation Counter (Perkin Elmer, Massachusetts, USA).

Skeletal muscle lysis

Frozen muscle tissue was powdered by grinding in a mortar and pestle chilled with liquid nitrogen and dry ice. To lyse powdered tissue, 200 μ L lysis buffer (4% (w/v) sodium deoxycholate, 100 mM Tris-HCl pH 8.5) was added followed by 10 s vortexing. Samples were then sonicated at 4 °C at 90% power using pulses of 2 s on, 5 s off for a total time of 1 min. Samples were then immediately boiled at 95 °C with 1,500 rpm shaking for 5 min and sonicated for a further 2 min (4 °C, 90% power, 5 s on and 5 s off) to ensure complete lysis. Lysate was then centrifuged at 20,000 xg for 5 min and 180 μ L supernatant was collected. Cysteine residues were reduced and alkylated

by adding 40 mM chloroacetamide and 10 mM TCEP at pH 7. Lysate was incubated for 5 min at 45 °C with 1,500 rpm shaking and then incubated for a further 40 min at room temperature without shaking.

Next, 800 µL 100% chloroform and 1,600 µL 100% methanol were added following 30 s sonication at 90% power. LC/MS grade water (800 µL) was added following 5 min mixing at 1,000 rpm. Lysate was centrifuged for 5 min at 2,000 xg to induce a phase separation. The majority of the aqueous phase (2,400 µL) was removed and 2,000 µL was reserved for ³H-2DG quantification. Next, 2,400 µL 100% methanol was added following 30 s mixing at 800 rpm and centrifugation at 2,000 xg for 5 min. The supernatant was discarded, and the protein pellet was air-dried for 5 min. Protein was reconstituted in 200 µL lysis buffer, sonicated at 60% power for 15 s using a tip-probe sonicator and boiled for 5 min in a thermomixer at 95 °C with 1,500 rpm shaking.

Determining ³H-2DG uptake into muscle tissue

Anion exchange chromatography was used to quantify ³H-p2DG, representing ³H-2DG that has been taken up by cells. For quantification of total (phosphorylated and unphosphorylated) ³H-2DG, 375 µL lysate aqueous phase was combined with 1,125 µL water and reserved. For quantification of unphosphorylated ³H-2DG, 1,000 µL lysate aqueous phase was added to 300 µL 37.5% (w/v) AG 1-X8 anion exchange resin (Bio-Rad, Hercules, CA, USA: 1401441) and washed with 3 mL water to remove p2DG. Liquid scintillation cocktail (3 mL) was then added to 1,500 µL total and unphosphorylated ³H-2DG solutions, and ³H-2DG was measured using a Tri-Carb 2810R Liquid Scintillation Counter. Unphosphorylated and total ³H-2DG scintillation counts were subtracted to quantify ³H-p2DG.

³H-2DG blood concentration at 1, 5, 7.5, and 10 min after injection was fit to an exponential curve $y = C_p(0)e^{-K_p t}$ where $C_p(0)$ represents the predicted initial tracer concentration (DPM/µL) and K_p represents the rate of tracer disappearance from the blood (1/min), to model the disappearance of ³H-2DG from the blood as it is taken up and trapped by peripheral tissues⁸⁰. $C_p(1)$ was removed when it was abnormally low ($C_p(1) < C_p(5)$, $C_p(5) - C_p(1) > 0.5 \times (C_p(5) - C_p(7.5))$), which likely indicates insufficient diffusion of circulating ³H-2DG into the tail vein. To account for different rates of blood ³H-2DG disappearance across mice, ³H-2DG uptake was calculated as a rate constant⁸⁰.

$$K_i = \frac{C_i(t)K_p}{C_p(0)(1 - e^{-K_p t})}$$

Where t represents the time after injection that the animal was sacrificed (min) and $C_i(t)$ represents the concentration of ³H-p2DG in the tissue harvested at time t (DPM/mg tissue).

Phosphoproteomics sample preparation

Phosphopeptides were isolated using the EasyPhos protocol²⁸ with minor modifications. Protein (C57Bl6J and NOD: 755 µg, BXH9 and BXD34: 511 µg, CAST: 364 µg) was digested into peptides by incubation in 1% (w/w) Trypsin and LysC on a thermomixer at 37°C with 1,500 rpm shaking for 14 h. Following digestion, 400 µL 100% isopropanol and 100 µL EasyPhos enrichment buffer (48% (v/v) TFA, 8 mM KH₂PO₄) were sequentially added with mixing (1,500 rpm, 30 s) after each addition. Lysate was centrifuged at 20,000 xg for 15 min to pellet insoluble material and transferred to a deep well plate. The EasyPhos protocol was then followed from step 12²⁸.

Liquid chromatography-tandem mass spectrometry (LC-MS/MS)

Enriched phosphopeptides in MS loading buffer (2% ACN, 0.3% TFA) were loaded onto in-house fabricated 55 cm columns (75 µM I.D.), packed with 1.9 µm C18 ReproSil Pur AQ particles (Dr. Maisch HPLC GmbH, Ammerbuch, DEU) with a Dionex U3000 HPLC (Thermo Fisher Scientific),

interfaced with an Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific). Column temperature was maintained at 60°C using a column oven (Sonation lab solutions, Biberach, DEU), and peptides were separated using a binary buffer system comprising 0.1% formic acid (buffer A) and 80% ACN plus 0.1% formic acid (buffer B) at a flow rate of 400 nL/min. A gradient of 3–19% buffer B was employed over 40 min followed by 19–41% buffer B over 20 min, resulting in approximately 1 h gradients. Peptides were analysed with one full scan (350–1,400 m/z, R = 120,000) at a target of $3e^6$ ions, followed by 48 data-independent acquisition MS/MS scans (350–1,022 m/z) with higher-energy collisional dissociation (target $3e^6$ ions, max injection time 22 ms, isolation window 14 m/z, 1 m/z window overlap, normalised collision energy 25%), and fragments were detected in the Orbitrap (R = 15,000).

MS raw data processing

Raw spectral data were analysed using Spectronaut (v16.0.220606.53000). Data were searched using directDIA against the Mouse UniProt Reference Proteome database (January 2022 release), with default settings (precursor and protein Qvalue cutoffs 0.01, Qvalue filtering, MS2 quantification), with “PTM localization” filtering turned on (threshold 0.5), and the inbuilt peptide collapse function.

Phosphoproteomics data processing

Phosphopeptide intensities were log2 transformed and median normalised. Non-class I phosphopeptides (localisation score ≤ 0.75) were then removed. Finally, for each phosphopeptide, outlier values were removed that had a log2 intensity < 5 and were > 6 log2 intensity units lower than the phosphopeptide median. Log2 fold changes between conditions were computed using condition medians.

Identifying insulin-regulated phosphopeptides

To allow comparison across conditions, phosphopeptides were filtered for those highly quantified in most strain-diet combinations. For a given phosphopeptide, this filtering was performed on two levels. Firstly, each of the 10 strain-diet combinations were retained if there were ≥ 3 insulin-stimulated values and ≥ 3 unstimulated values. Then, the phosphopeptide itself was retained if ≥ 8 strain-diet combinations had passed the previously filtering. Phosphopeptides were then fit to a three-way ANOVA with all interaction terms (“aov” in the R package “stats”) and an F-test was performed assessing the main effect of insulin stimulation. To correct for multiple hypothesis testing p-values were converted into q-values (R package “qvalue”⁸¹). The log2(insulin/unstimulated) fold change with the greatest magnitude across strain-diet combinations was then calculated (max log2(insulin/unstimulated)). Phosphopeptides were considered “insulin-regulated” when $q < 0.05$ and if insulin increased or decreased phosphorylation by > 1.5 -fold in at least one strain-diet combination (max log2(insulin/unstimulated) > 0.58 or < -0.58).

Calculation of insulin response values

For all phosphopeptides the distribution of “insulin responses” in each strain-diet combination was calculated. Specifically, within each strain-diet combination all insulin-stimulated values were normalised by subtracting the unstimulated median.

Identifying strain and diet effects

Strain effects

For each insulin-regulated phosphopeptide a one-way ANOVA was performed modelling the insulin response as a function of mouse strain within the CHOW diet. p-values were converted to q-values. For significant phosphopeptides ($q < 0.05$), t-tests were performed comparing the insulin response of C57Bl6J to each of the other four strains. t-test p-values were converted to q-values and considered significant when $q < 0.05$. To ensure that significant differences between a strain and C57Bl6J were of a meaningful magnitude, the strain's $\log_2(\text{insulin}/\text{unstimulated})$ was compared to the C57Bl6J $\log_2(\text{insulin}/\text{unstimulated})$. In general, if the absolute difference between the two was greater than 0.58 this was accepted. However, this threshold was decreased for phosphopeptides with weaker insulin regulation. Specifically, the difference was retained if it passed the following filtering:

$\text{Strain } \log_2(\text{insulin}/\text{unstimulated}) > f(\text{C57Bl6J } \log_2(\text{insulin}/\text{unstimulated}))$
or
 $\text{Strain } \log_2(\text{insulin}/\text{unstimulated}) < g(\text{C57Bl6J } \log_2(\text{insulin}/\text{unstimulated}))$,
where $f(x)$ and $g(x)$ are defined as:

$$f(x) = \begin{cases} x + 0.58, & \text{if } x \geq 2 \times 0.58 \\ \frac{5}{4}x + \frac{1}{2} \times 0.58, & \text{if } 0 \leq x < 2 \times 0.58 \\ x + \frac{1}{2} \times 0.58, & \text{if } -\frac{1}{2} \times 0.58 \leq x < 0 \\ \frac{4}{5}x + \frac{2}{5} \times 0.58, & \text{if } -3 \times 0.58 \leq x < -\frac{1}{2} \times 0.58 \\ x + 0.58, & \text{if } x < -3 \times 0.58 \end{cases}$$

$$g(x) = \begin{cases} x - 0.58, & \text{if } x \geq 3 \times 0.58 \\ \frac{4}{5}x - \frac{2}{5} \times 0.58, & \text{if } \frac{1}{2} \times 0.58 \leq x < 3 \times 0.58 \\ x - \frac{1}{2} \times 0.58, & \text{if } 0 \leq x < \frac{1}{2} \times 0.58 \\ \frac{5}{4}x - \frac{1}{2} \times 0.58, & \text{if } -2 \times 0.58 \leq x < 0 \\ x - 0.58, & \text{if } x < -2 \times 0.58 \end{cases}$$

An insulin-regulated phosphopeptide was considered to have a “Strain effect” if the insulin response in at least one strain was different to C57Bl6J using the q-value and $\log_2$ fold-change criteria described above.

Uniform diet and StrainxDiet effects

For each insulin-regulated phosphopeptide a two-way ANOVA was performed modelling the insulin response as a function of strain, diet, and their interaction. The p-values for the Diet and StrainxDiet terms were converted to q-values. Whenever the StrainxDiet term was significant ($q < 0.05$), additional tests were performed to identify specific strains in which the insulin response differed between CHOW and HFD. If the StrainxDiet term was not significant but the Diet term was significant, a separate filtering procedure was performed.

When the StrainxDiet term was significant, t-tests were performed to compare the CHOW insulin response to the HFD insulin response within each strain. When a t-test was significant ($q < 0.05$), the $\log_2(\text{insulin}/\text{unstimulated})$ filtering procedure described for “Strain effects” was applied comparing CHOW and HFD fold changes. Insulin-regulated phosphopeptides were considered to have a “StrainxDiet effect” if there was a difference between CHOW and HFD in at least one strain.

When only the Diet term was significant, the $\log_2(\text{insulin/unstimulated})$ filtering procedure described for “Strain effects” was applied, comparing the mean $\log_2(\text{insulin/unstimulated})$ across strains within CHOW, to the mean across HFD. Insulin-regulated phosphopeptides that passed this filter were considered to have a “Uniform diet effect”.

Curated insulin signalling subnetwork

A subnetwork of insulin-regulated phosphosites was curated by compiling all sites on proteins from a previously published knowledge pathway-derived list of canonical insulin signalling proteins²². Several phosphosites and proteins that were not detected as insulin regulated were included due to their importance in the insulin signalling pathway. Additionally, all *in vivo* substrates of canonical insulin-regulated kinases (Akt, mTOR, AMPK, Raf, Mek1/2, Erk1/2, p90RSK/Rsk2, p70S6K, Pdk1, INSR) annotated in PhosphositePlus were included³³. Annotations from orthologous phosphosites were pooled across species using PhosphositePlus Site Group IDs. Phosphosite regulatory roles from PhosphositePlus were indicated after manual validation by literature search. Proteins were assigned to functional groups (e.g. mRNA processing, lipid metabolism) based on their Uniprot descriptions.

Kinase substrate enrichment analysis

Kinase-substrate annotations were collated from PhosphositePlus and mapped into phosphoproteomics data using Site Group IDs. Only annotations supported by *in vivo* evidence were used. Annotations for kinase isoforms (e.g. Akt1, Akt2, Akt3) were merged. Substrate annotations for GSK3 were supplemented with a recent list of 274 putative GSK3 substrates determined by phosphoproteomics and motif analysis³². Autophosphorylation sites and promiscuous phosphosites targeted by ≥ 4 kinases were removed. Kinase substrate enrichment analysis (KSEA) was then performed with the “ksea” function from the R package “ksea”⁴⁰ (version: 0.1.2) using insulin response data and 1,000 permutations to determine empirical p-values. Only phosphopeptides quantified in $\geq 50\%$ of samples and with ≥ 1 insulin response value in all strain-diet combinations were used. In each sample kinases with < 5 quantified substrates were excluded, and only kinases with significant enrichment ($p < 0.05$) in ≥ 5 samples were used in subsequent analysis. To identify Strain effects on kinase activity, one-way ANOVAs were performed on CHOW KSEA enrichment scores. To identify Uniform diet or StrainxDiet effects, two-way ANOVAs were performed on KSEA enrichment scores testing the effects of strain, diet, and their interaction. p-values were adjusted by the Benjamini-Hochberg procedure.

Insulin signalling subnetwork analysis

Weighted gene correlation network analysis (WGCNA^{42,43}) was performed with the “blockwiseModules” function from the R package “WGCNA” (version 1.71) using the insulin response values of all insulin-regulated phosphopeptides. Default parameters were used except for power = 3 (determined as recommended in⁴²), deepSplit = 3, minModuleSize = 15, reassignThreshold = 0, and mergeCutHeight = 0.25. Subnetwork eigengenes were extracted and termed “eigenpeptides”. One-sided fisher’s exact tests were performed to assess the enrichment of Gene Ontology (GO) Biological Processes, GO Cellular Compartments (R package “org.Mm.eg.db” version 3.15.0⁸²), and kinase substrates in each subnetwork relative to the entire phosphoproteome. Only pathways containing three or more subnetwork phosphoproteins were tested. Kinase substrate enrichment was performed using the same annotations as KSEA. P-values were adjusted within each analysis by the Benjamini-Hochberg procedure. Subnetwork phosphopeptides were mapped into insulin signalling temporal clusters defined in our previous study of insulin signalling dynamics¹⁰, using PhosphositePlus Site Group IDs. The timepoint at which each cluster appeared to reach its maximum insulin-stimulated value was used as a measure of insulin response speed.

Glucose uptake correlations

For each insulin-regulated phosphopeptide, Pearson's correlation tests were performed to assess the linear association between ^3H -2DG uptake in insulin-stimulated mice and phosphopeptide insulin response values or unnormalized insulin-stimulated log2 intensity. Phosphopeptides were considered correlated with ^3H -2DG uptake when $q < 0.1$ and their Pearson's correlation coefficient was of substantial magnitude ($r > 0.35$ or $r < -0.35$). Pearson's correlation tests were also performed comparing insulin-stimulated ^3H -2DG uptake to KSEA enrichment scores in individual mice or using the median in each strain-diet combination.

Cell culture

GLUT4-HA-L6 myoblasts⁸³ were grown in α -MEM supplemented with 10% fetal bovine serum in a humidified chamber at 37 °C, 10% CO₂. Differentiation was induced by changing media to α -MEM supplemented with 2% horse serum for 5 days.

Pfkfb3 overexpression

Platinum-E (Plat-E) retroviral packaging cells were grown to confluency and transfected with 10 μg total DNA: either pBabe puromycin empty vector, pBabe puromycin Pfkfb3 or pWZL neomycin HA-GLUT4. After 48 h retroviral media was collected and passed through a 0.45 μm filter. L6 myotubes were grown to confluence and retrovirally transfected with 2 mL of HA-GLUT4 neomycin viral media in the presence of 10 $\mu\text{g}/\text{mL}$ polybrene. The following morning, cells were split into growth media containing neomycin (800 $\mu\text{g}/\text{mL}$) and passaged until only successfully transfected cells remained. These cells were then grown to confluence and retrovirally transfected again with 2 mL of either empty vector puromycin viral media or Pfkfb3 puromycin viral media in the presence of 10 $\mu\text{g}/\text{mL}$ of polybrene. The following morning, cells were split into growth media containing both neomycin (800 $\mu\text{g}/\text{mL}$) and puromycin (2 $\mu\text{g}/\text{mL}$) and passaged until only successfully transfected cells remained in culture.

Extracellular acidification rate

The extracellular acidification rate (ECAR) of GLUT4-HA-L6 cells myotubes was measured using Seahorse XFp miniplates and a Seahorse XF HS Mini Analyzer (Seahorse Bioscience, Copenhagen, Denmark) as previously described⁸⁴. Cells incubated in palmitate or BSA control were washed twice with Krebs-Ringer Bicarbonate Buffer (Sigma, K4002) and once with standard cell culture media without bicarbonate (XF-DMEM, pH 7.4). Cells were then incubated in XF-DMEM without glucose at 37°C for 1 h in a non-CO₂ incubator, followed by assaying in the XFp Analyzer. ECAR was measured after a 12-minute equilibration period followed by mix/wait/read cycles of 3/0/3 min. After stabilizing the baseline rates, compounds were injected to reach a final concentration of: 10 mM glucose, 5 $\mu\text{g}/\text{mL}$ oligomycin, and 50 mM 2-deoxyglucose (2-DG), allowing estimation of glucose-driven glycolysis (glucose ECAR - basal ECAR), glycolytic capacity (oligomycin ECAR - 2DG ECAR), and non-glycolytic acidification (equal to 2DG ECAR). Data were normalized to protein concentration and presented as a percentage of total ECAR.

2DG uptake in GLUT4-HA-L6 myotubes

2-deoxyglucose (2DG) uptake into GLUT4-HA-L6 myotubes was performed as previously described with modifications^{83,85}. Cells were incubated overnight in α MEM supplemented with either BSA-coupled 125 μM palmitic acid or BSA vehicle control before being washed 3x with 37 °C HEPES-buffered saline (HBS). Cells were then incubated in HBS supplemented with 10 μM unlabelled 2-deoxyglucose and either 0 or 100 nM insulin at 37°C for 15 min. Cells were then incubated for a further 5 min following the addition of 0.5 $\mu\text{Ci}/\text{mL}$ [^3H]-2-deoxyglucose in HBS. Cells were then washed on ice 5x with ice-cold PBS and lysed in 1 M NaOH. For non-specific background

uptake, 1 well per condition was pre-treated with cytochalasin B. Counts were determined by Perkin Elmer Quantulus GCT Liquid Scintillation Counter (Perkin Elmer, Waltham, MA, USA). 2DG uptake was expressed relatively to protein concentration as determined by bicinchoninic acid (BCA) assay after neutralisation with 1 M HCl and subtraction of non-specific uptake.

Immunoblotting

GLUT4-HA-L6 myotubes were incubated overnight (16 h) in either BSA-conjugated 125 μ M palmitate or BSA vehicle control. Cells were optionally treated with insulin as described above, and then washed in ice-cold PBS and lysed by scraping directly into 55 °C Laemmli sample buffer with 10 % (tris 2-carboxyethyl phosphine; TCEP). Samples were sonicated for 24 s (3s on/3s off) and heated at 65 °C for 5 minutes. Samples were then resolved by SDS-PAGE as previously described¹⁸, transferred onto PVDF membranes and blocked in TBS-T (0.1% Tween in Tris-buffered saline) containing 5% skim milk for 1 h. Membranes were then washed 3 x 10 min in TBS-T and incubated overnight in primary antibodies against Pfkfb3 (Proteintech Group; 13763-1-AP) and α -tubulin (Cell Signalling Technologies #2125; diluted 1:1000). Insulin-stimulated cells were additionally incubated in primary antibodies against pT308 Akt (Cell Signaling Technologies #2965; diluted 1:1000), pS473 Akt (Cell Signaling Technologies #9271; diluted 1:1000), total pan-Akt (Cell Signaling Technologies #9272; diluted 1:1000), pS21/S9 GSK3 α/β (Cell Signaling Technologies #9327; diluted 1:1000), pT246 PRAS40 (Cell Signaling Technologies #13175; diluted 1:1000), total PRAS40 (Cell Signaling Technologies #2691, diluted 1:1000, and 14-3-3 (Santa Cruz #sc-1657, diluted 1:5000). The following day membranes were washed 3 x 10 min in TBS-T and incubated for 1 h in species-appropriate fluorescent antibodies. Imaging and densitometry were performed using LI-COR Image Studio.

Table S1: Muscle phosphoproteomics (Page 1 “01_quantification”) Normalized LFQ intensities of class I phosphopeptides. (Page 2 “02_analysis”) Statistical analysis of phosphoproteome data.

Table S3: Insulin signalling subnetworks WGCNA-derived subnetworks of insulin-regulated phosphopeptides.

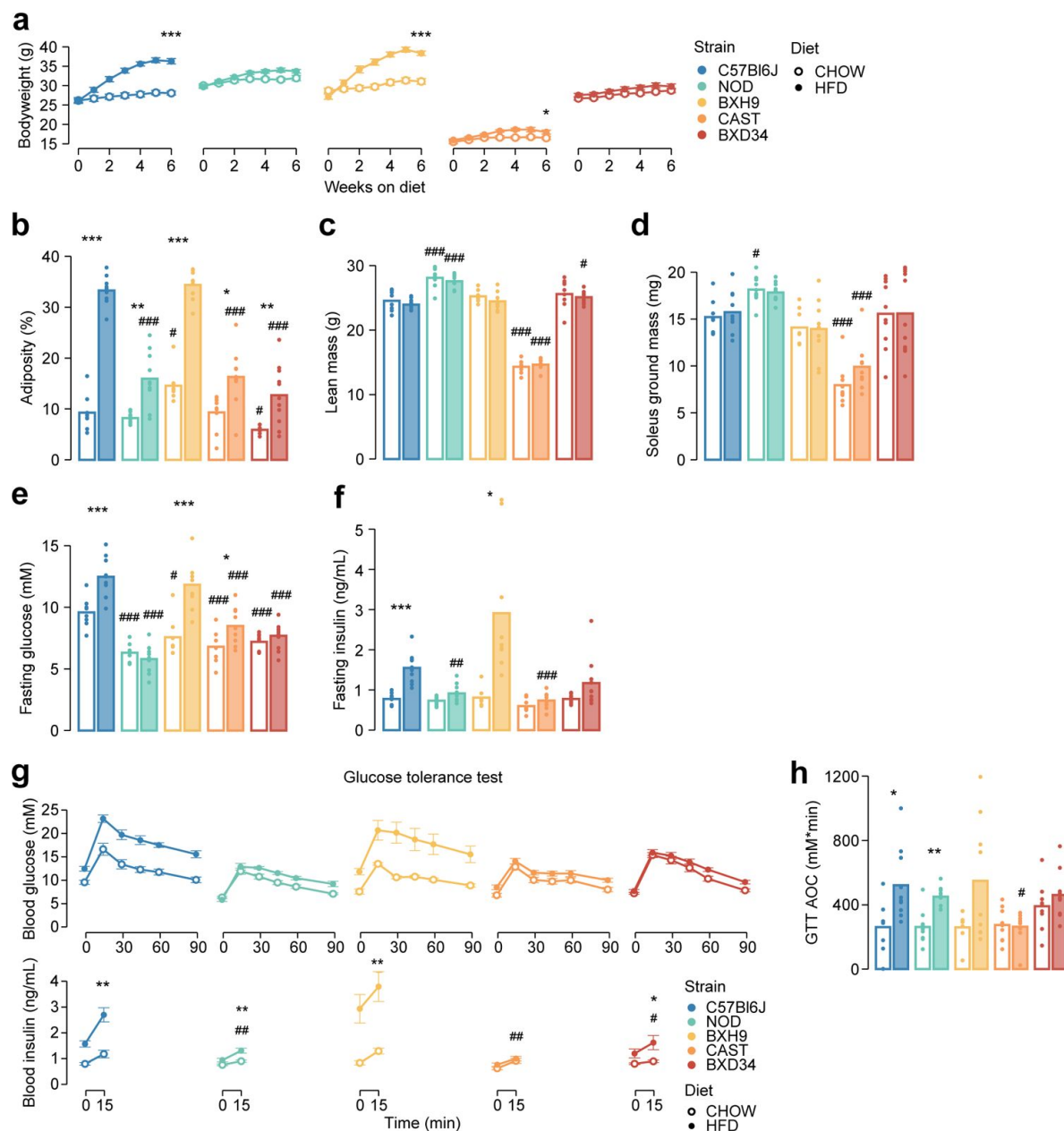


Figure S1

Genetics and diet alter morphometric and metabolic phenotypes. Related to Fig. 1

a) Mouse bodyweight was measured during a six-week diet regimen. Two-sided t-tests were performed to compare HFD to CHOW within each strain after six weeks, following Benjamini-Hochberg p-value adjustment (*). **b-d)** Measurement of **b)** adiposity, **c)** lean mass, **d)** ground soleus mass, **e)** fasting blood glucose, and **f)** fasting blood insulin at the end of the diet regimen. **g)** At the end of the diet regimen a glucose tolerance test was performed. **h)** The area of the blood glucose curve (GTT AOC) was calculated. In **b-h)**, two-sided t-tests were performed to compare HFD to CHOW within each strain (*) or to compare each strain to C57Bl6J within either diet (#). P-values were adjusted by the Benjamini-Hochberg procedure. Error bars indicate SEM. In **g)** t-tests were only performed on 15-minute blood insulin levels. No comparisons across strains on CHOW were significant. $n = 8-11$ biological replicates. */#: $0.01 \leq p < 0.05$, **/##: $0.001 \leq p < 0.01$, ***/###: $p < 0.001$

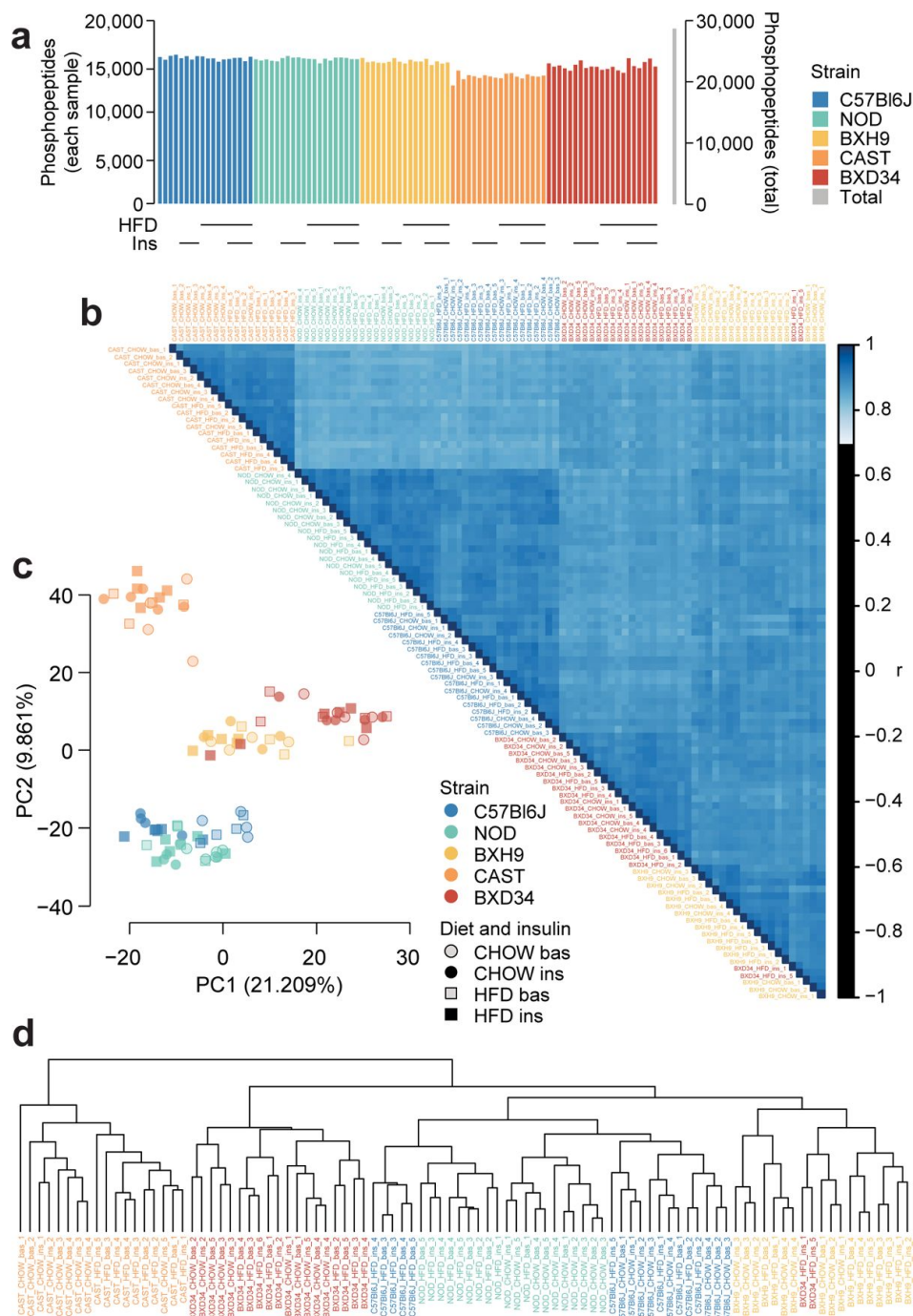


Figure S2

Quality control analysis of phosphoproteomics data. Related to Fig. 1. a) The number of unique class I phosphopeptides quantified in each sample and in total. b) Pearson's correlation was performed between each pair of samples. Samples are ordered by hierarchical clustering. c) Principal component analysis was performed on the phosphoproteome. The first two principal components (PC1 and PC2) are plotted for each sample and the percentage of overall variance explained by each principal component is indicated. "bas": unstimulated, "ins": insulin-stimulated. d) Hierarchical clustering was performed on all samples.

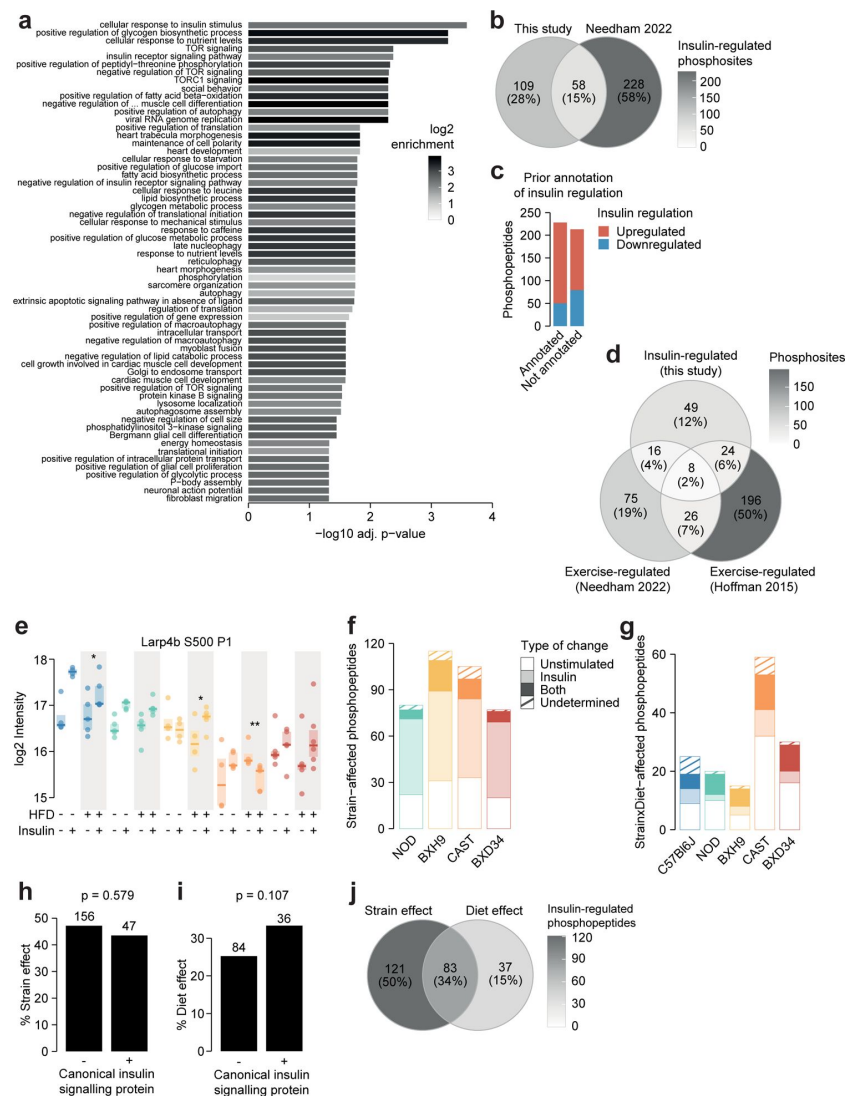


Figure S3

Characterisation of the insulin-regulated phosphoproteome.

Related to **Fig. 1** [1](#) [2](#) [3](#). **a**) The enrichment of GO biological processes in genes containing insulin-regulated phosphopeptides relative to the entire phosphoproteome (one-sided Fisher's exact test, Benjamini-Hochberg p-value adjustment). Only significant pathways are shown (adj. p < 0.05). The pathway "negative regulation of vascular associated smooth muscle cell differentiation" is abbreviated. **b**) The number of phosphosites regulated by insulin in this study or a previous phosphoproteomic study of human skeletal muscle¹. Only phosphosites quantified in both studies were considered. **c**) The number of insulin-regulated phosphopeptides with prior annotation of insulin regulation in the PhosphositePlus database². **d**) The number of phosphosites regulated by insulin this study or regulated by exercise in two human phosphoproteomics studies^{1,3}. Only phosphosites quantified in all three studies were considered. **e**) A phosphopeptide where HFD-feeding enhanced insulin responses in BXH9 but suppressed insulin responses in C57Bl6j and CAST. A two-way ANOVA was performed on insulin response values followed by two-sided t-tests comparing HFD to CHOW within each strain (q-values: *). **f**) Phosphopeptides with a Strain effect were examined to determine whether the effect was due to altered unstimulated phosphorylation ("Unstimulated"; Strain/C57Bl6j fold change > 1.3 in unstimulated samples), altered insulin-stimulated phosphorylation ("Insulin"; Strain/C57Bl6j fold change > 1.3 in insulin-stimulated samples), or both ("Both"). A proportion of phosphopeptides passed neither of these filters ("Undetermined"). **g**) The same analysis was performed on StrainxDiet-affected phosphopeptides, using the HFD/CHOW fold changes in either unstimulated or insulin-stimulated samples for each strain. **h-i**) The percentage of **h**) Strain effects and **i**) Diet effects (Uniform diet or StrainxDiet effect) among canonical or noncanonical insulin signalling proteins. P-values indicate two-sided Fisher's exact tests. The number of phosphopeptides in each group is shown. **j**) The overlap of Strain and Diet effects.

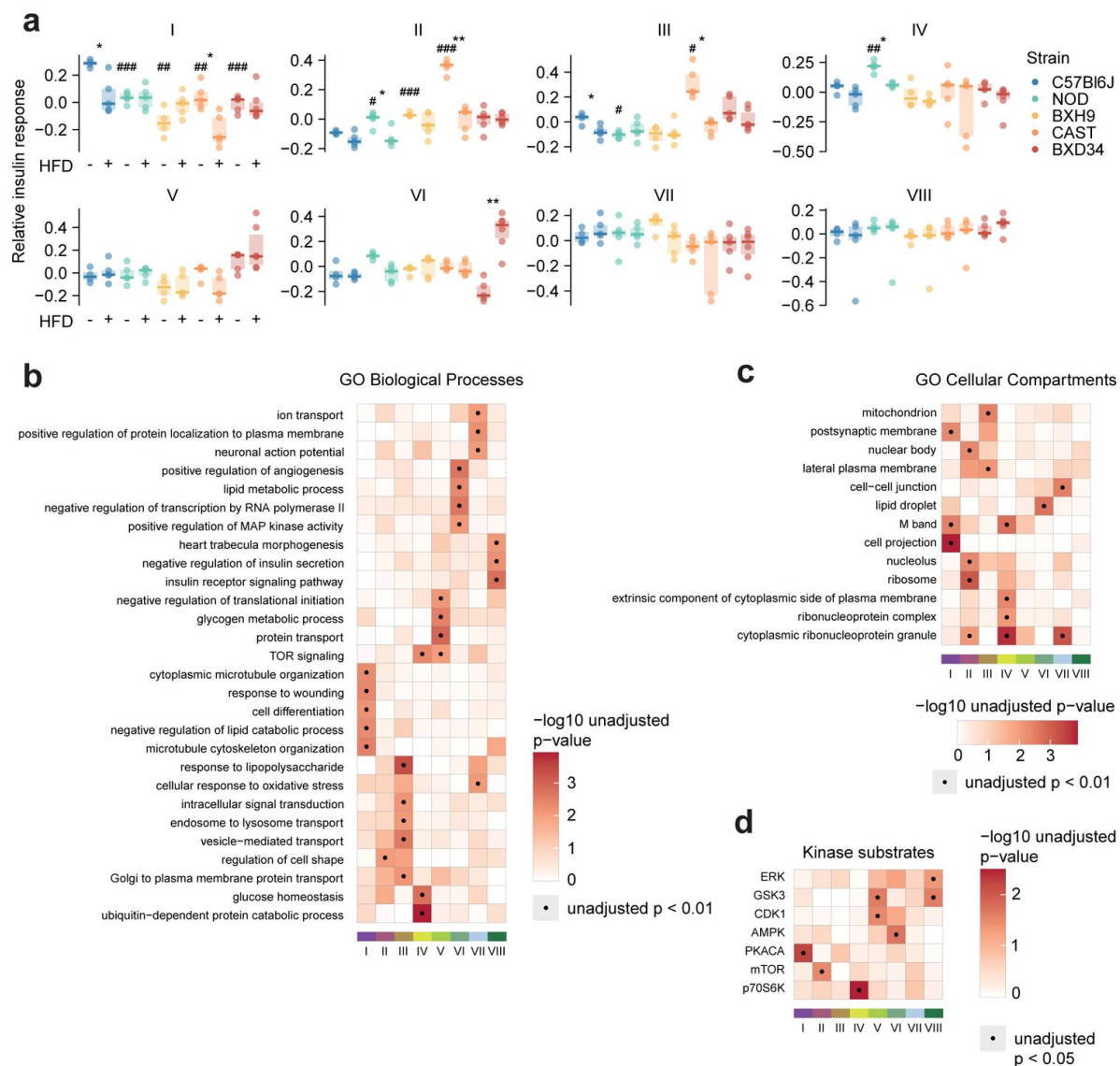


Figure S4

Characterising insulin signalling subnetworks. Related to Fig. 5. a) The eigenpeptides of each WGCNA-derived subnetwork. ANOVAs were performed on CHOW values following two-sided t-tests comparing each strain to C57Bl6J (Benjamini-Hochberg-adjusted p-values: #), and two-way ANOVAs were performed on all values followed by two-sided t-tests comparing HFD to CHOW within each strain (adjusted p-values: *). b-d) Rank-based enrichment of GO biological pathways, GO cellular compartments, and PhosphositePlus kinase substrates using phosphopeptide module membership scores with the “geneSetTest” function from the R package “limma”. Membership score is defined as the absolute Pearson’s correlation coefficient between a phosphopeptide’s insulin response and the subnetwork’s eigenpeptide. */#: $0.01 \leq p < 0.05$, **/##: $0.001 \leq p < 0.01$, ***/###: $p < 0.001$

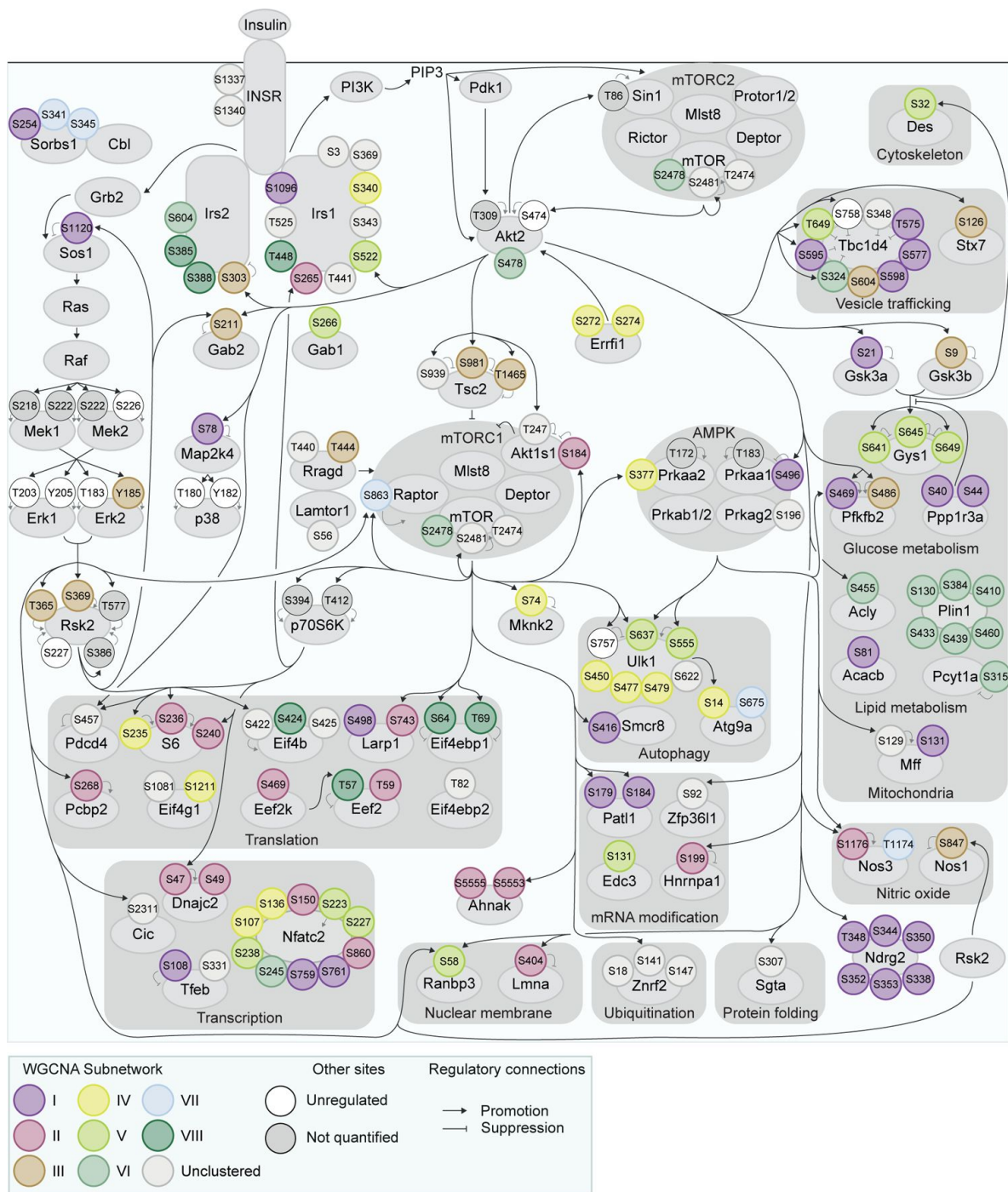


Figure S5

Organisation of insulin signalling subnetworks.

Related to [Fig. 5](#). The curated insulin signalling network displayed in [Fig. 3](#) was annotated with WGCNA subnetworks from [Fig. 5](#).

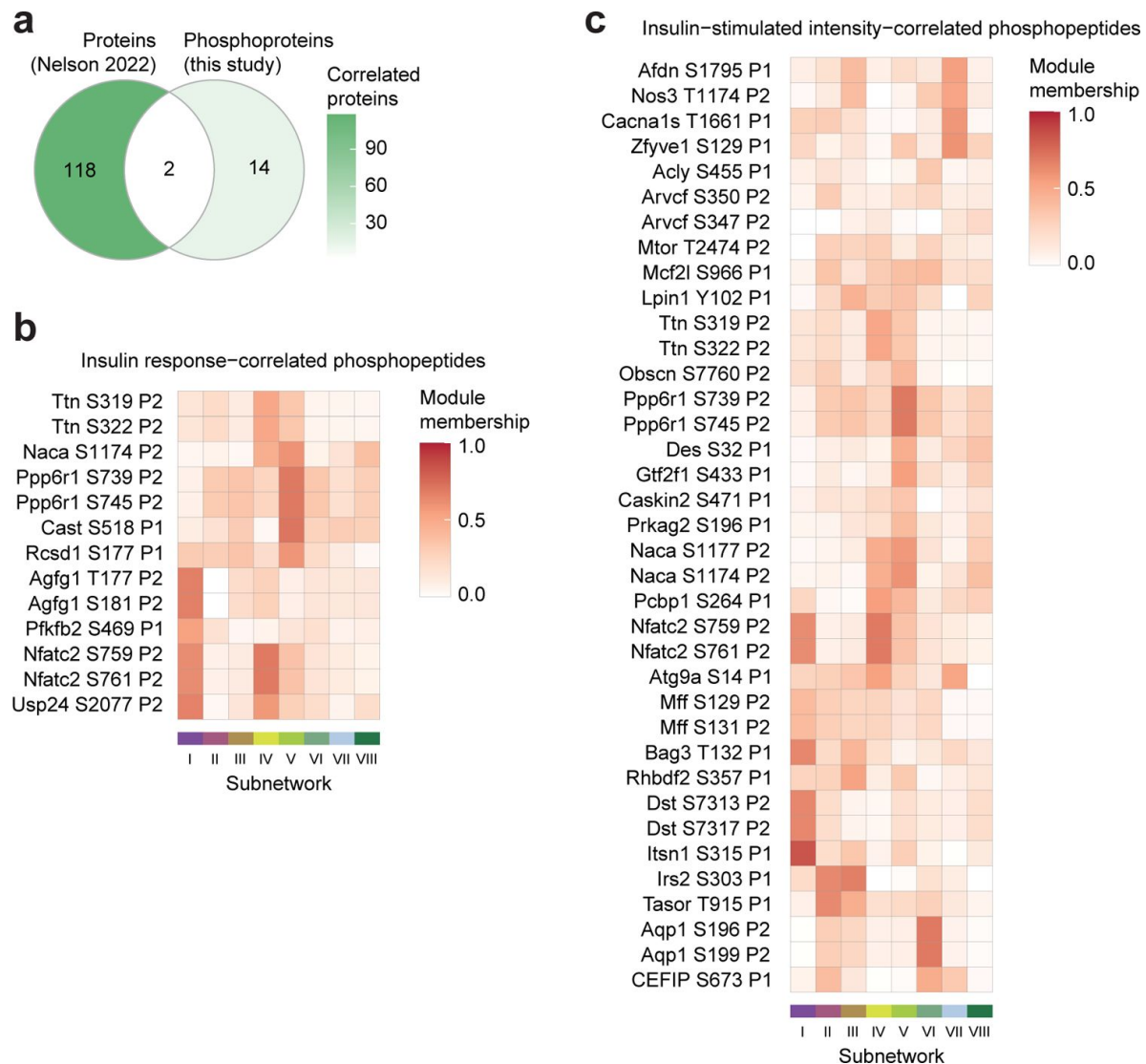


Figure S6

Characterisation of glucose-uptake correlated phosphosites. Related to Fig. 6. a) The intersection of proteins correlated with insulin-stimulated glucose uptake in the soleus of seven inbred mouse strains fed CHOW or HFD⁴ ($p < 0.1$, $r > 0.35$ or < -0.35), with proteins containing glucose-uptake correlated phosphopeptides in this study. Only (phospho)proteins quantified in both studies are shown. b-c) The subnetwork membership scores for glucose-uptake correlated phosphopeptides using b) insulin-stimulated phosphopeptide intensity, or c) phosphopeptide insulin response values. Membership score is defined as the absolute Pearson's correlation coefficient between a phosphopeptide's insulin response and the subnetwork's eigenpeptide.

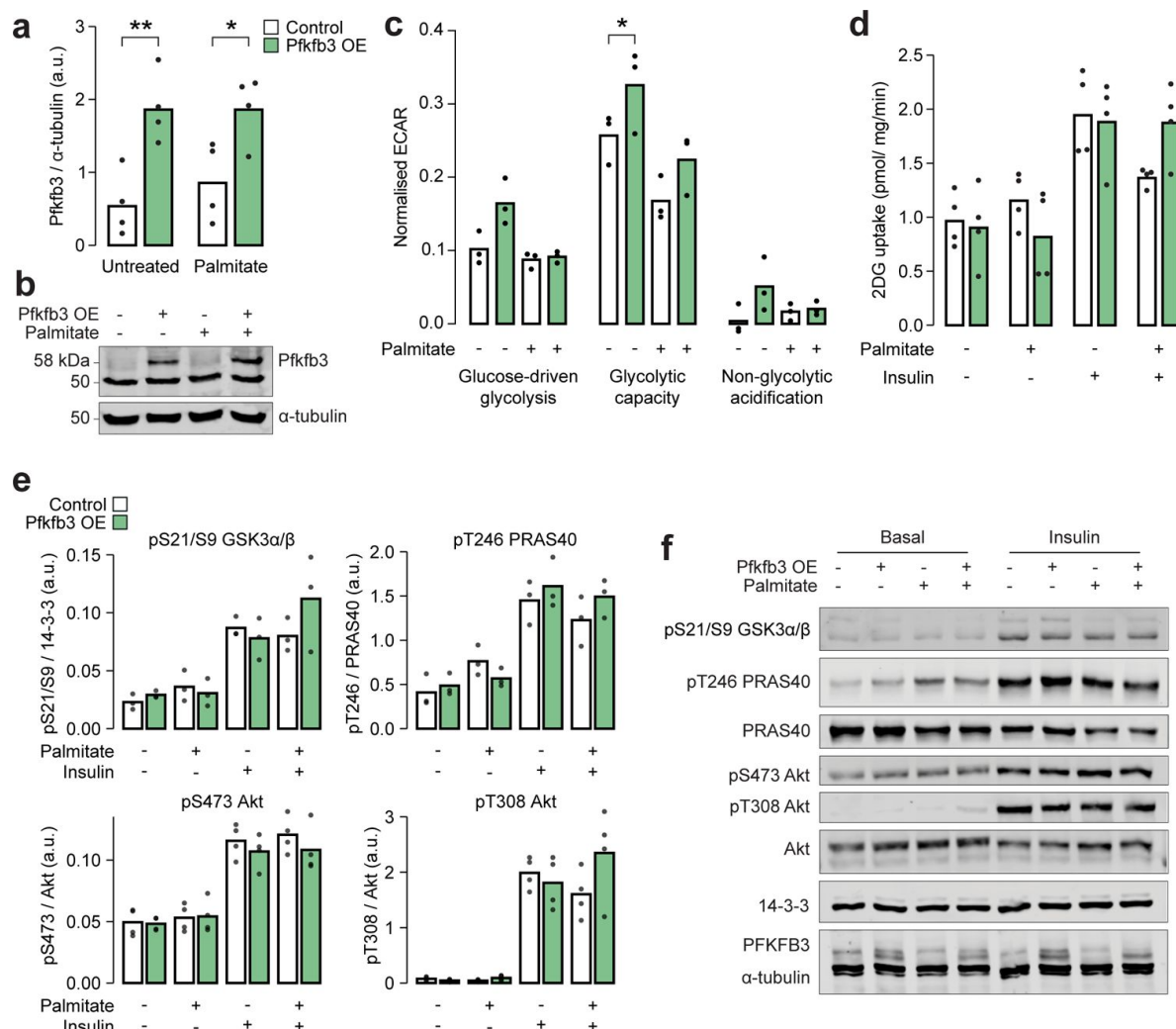


Figure S7

Overexpression of Pfkfb3 enhances glycolytic capacity and reverses palmitate-induced insulin resistance. Related to Fig. 6. a) Quantification and b) representative blot for immunoblotting of Pfkfb3 in L6-GLUT4-HA myotubes with or without Pfkfb3 overexpression, treated with palmitate (125 μ M, 16 h) or BSA vehicle control. Two-way ANOVA was performed followed by Šidák's post-hoc tests assessing the effect of Pfkfb3 overexpression (*). $n = 4$ biological replicates. c) Extracellular acidification rate (ECAR) in L6-GLUT4-HA myotubes treated with glucose (10 mM, "Glucose-driven glycolysis"), oligomycin (5 μ g/mL, "Glycolytic capacity"), or 2-deoxyglucose (50mM, "Non-glycolytic acidification"). A two-way ANOVA was performed followed by Tukey's posthoc tests comparing conditions within each of the three treatments (*). Not all significant comparisons are shown. $n = 3$ biological replicates. d) Unstimulated and insulin-stimulated glucose uptake (100 nM insulin, 20 min) in L6-GLUT4-HA myotubes. Insulin/unstimulated fold changes are shown in Fig. 6h. e) Quantification and f) representative blot for immunoblotting of insulin signalling phosphosites in unstimulated or insulin-stimulated (100 nM, 20 min) L6-GLUT4-HA myotubes. */#: $0.01 \leq p < 0.05$, **/##: $0.001 \leq p < 0.01$, ***/###: $p < 0.001$

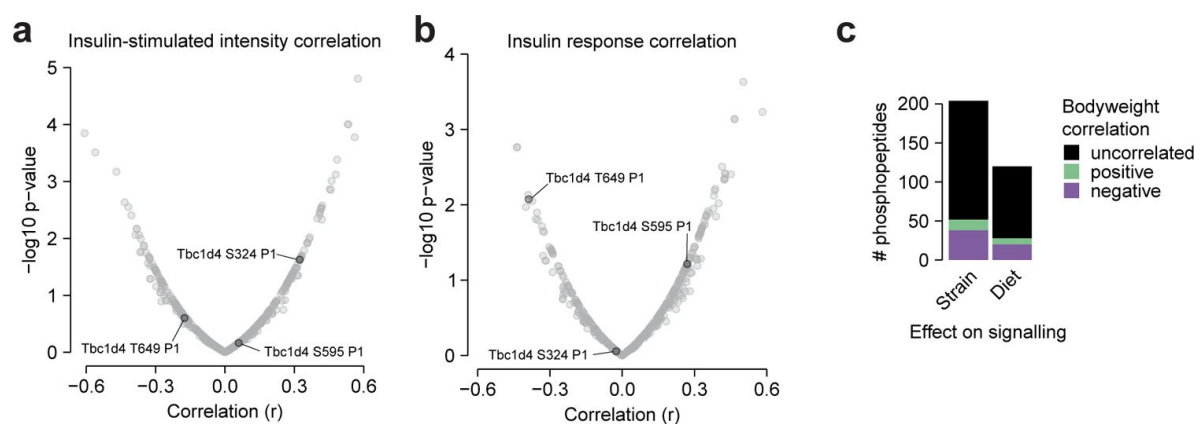


Figure S8

Additional glucose uptake correlation analysis. Related to Discussion. a-b) The correlation of insulin-stimulated glucose uptake with insulin-regulated phosphopeptides using a) insulin-stimulated phosphopeptide intensity, or b) phosphopeptide insulin response values, as displayed in Fig. 6. Canonical regulatory phosphosites on Tbc1d4 are indicated. The fourth canonical regulatory site S758 was not analysed due to insufficient quantification (quantified in 4/94 samples). c) The correlation of mouse bodyweight with the insulin responses of Strain or Diet-affected phosphopeptides either in CHOW-fed mice (Strain effects) or across both diets (Diet effects). The number of positively correlated (Pearson's correlation, $q < 0.1$, $r > 0.35$) and negatively correlated ($q < 0.1$, $r < -0.35$) phosphopeptides is shown.

<i>Phosphosite details</i>				<i>Exercise regulation</i>	
PhosphositePlus					
Site	Group ID	Gene	Uniprot	Phosphosite	Hoffman 2015 Needham 2022
	450254	Eef2	P58252	T59	+ +
	448040	Eef2	P58252	T57	+ +
	470754	Mff	Q6PCP5	S131	+ +
	470753	Mff	Q6PCP5	S129	+ +
	453632	Larp1	Q6ZQ58	S498	+ +
	455925	Tbc1d4	Q8BYJ6	S324	+ +
	4762876	Svil	Q8K4L3	S300	+ +
	448238	Gys1	Q9Z1E4	S645	+ +
	480145	Tns1	E9Q0S6	S1054	+ +
	3176152	Ulk1	O70405	S450	+ +
	448064	Rps6ka3	P18654	S369	+ +
	469782	Dnajc2	P54103	S47	+ +
	486522	Pcbp1	P60335	S264	+ +
	448094	Rps6	P62754	S240	+ +
	448093	Rps6	P62754	S236	+ +
	447594	Mapk1	P63085	Y185	+ +
	453577	Patl1	Q3TC46	S179	+ +
	447527	Eif4ebp1	Q60876	S64	+ +
	482918	Pcbp2	Q61990	S268	+ +
	4731870	Speg	Q62407	S542	+ +
	14570198	Atg9a	Q68FE2	S675	+ +
	455922	Tbc1d4	Q8BYJ6	S348	+ +
	471364	Lpin1	Q91ZP3	S923	+ +
	448239	Gys1	Q9Z1E4	S649	+ +
	4693547	Ttn	A2ASS6	S322	+ +
	480195	Usp24	B1AY13	S1296	+ +
	10782556	Usp24	B1AY13	S1282	+ +
	483808	Akap13	E9Q394	S2692	+ +
	3208338	Akap1	O08715	S103	+ +
	3831732	Tom1	O88746	S376	+ +
	2048085	Rab12	P35283	S20	+ +
	447655	Map2k4	P47809	S78	+ +
	4720237	Lmod2	Q3UHZ5	T391	+ +
	450706	Hspb6	Q5EBG6	S16	+ +
	4726016	Tom1l2	Q5SRX1	S394	+ +
	4716961	Slc20a2	Q80UP8	S316	+ +
	17544627	Lmod1	Q8BVA4	S511	+ +
	11180047	Tbc1d4	Q8BYJ6	S604	+ +
	455921	Tbc1d4	Q8BYJ6	T649	+ +
	3195560	Golga4	Q91VW5	S41	+ +
	485723	Plekhf2	Q91WB4	S226	+ +
	4772204	Ehbp1l1	Q99MS7	S1460	+ +
	470067	Akt1s1	Q9D1F4	T247	+ +
	452430	Plec	Q9QXS1	S4620	+ +

Table S2

Overlap between insulin and exercise-regulated phosphosites Insulin-regulated phosphosites from this study that were additionally regulated by exercise either in Needham et al.¹ or Hoffman et al.³.

451009 Ndr2	Q9QYG0	T348	+
468257 Ndr2	Q9QYG0	S338	+
468259 Ndr2	Q9QYG0	S344	+
447886 Nos1	Q9Z0J4	S847	+

Table S2 (continued)

Kinase	Correlating all values		Correlating Strain-Diet medians	
	r	p	r	p
SGK	0.026282557	0.85773407	-0.114513813	0.752760165
Aur	0.181856398	0.221176679	0.446186208	0.196168185
p90RSK	0.042637179	0.771137201	0.392619919	0.261741297
aPKC	0.020857899	0.886881337	0.059269839	0.870801728
GSK3	-0.023496391	0.872683503	0.200939915	0.577765996
CDK5	0.092355979	0.527940053	0.136553891	0.706796386
P38	0.186315084	0.199914602	0.202333014	0.575075383
CDK1	0.134654975	0.356287427	0.245970422	0.493328434
CK2	0.062152944	0.671383691	-0.206449396	0.567152858

Table S4

Association of kinase enrichment with insulin-stimulated glucose uptake
earson's correlation of KSEA enrichment scores with insulin-stimulated
glucose uptake for all kinases with Strain or Diet effects. Correlation was
performed on all values or on the medians of each Strain-Diet combination.

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Reviewer #1 (Public Review):

The authors focused on genetic variability in relation to insulin resistance. They used genetically different lines of mice and exposed them to the same diet. They found that genetic predisposition impacts the overall outcome of metabolic disturbances.

- <https://doi.org/10.7554/eLife.89212.2.sa2>

Reviewer #2 (Public Review):**Summary:**

In the present study, van Gerwen et al. perform deep phosphoproteomics on muscle from saline or insulin-injected mice from 5 distinct strains fed a chow or HF/HS diet. The authors follow these data by defining a variety of intriguing genetic, dietary or gene-by-diet phosphor-sites which respond to insulin accomplished through application of correlation analyses, linear mixed models and a module-based approach (WGCNA). These findings are supported by validation experiments by intersecting results with a previous profile of insulin-responsive sites (Humphrey et al, 2013) and importantly, mechanistic validation of Pfkfb3 where overexpression in L6 myotubes was sufficient to alter fatty acid-induced impairments in insulin-stimulated glucose uptake. To my knowledge, this resource provides the most comprehensive quantification of muscle phospho-proteins which occur as a result of diet in strains of mice where genetic and dietary effects can be quantifiably attributed in an accurate manner. Utilization of this resource is strongly supported by the analyses provided highlighting the complexity of insulin signaling in muscle, exemplified by contrasts to the "classically-used" C57BL6/J strain. As it stands, I view this exceptional resource as comprehensive with compelling strength of evidence behind the mechanism explored. I raised several comments in the last round of assessment but all of them have now been thoughtfully addressed.

Strengths: Generation of a novel resource to explore genetic and dietary interactions influencing the phospho-proteome in muscle. This is accompanied by elegant application of in silico tools to highlight the utility

Weaknesses: none noted

- <https://doi.org/10.7554/eLife.89212.2.sa1>

Reviewer #3 (Public Review):**Summary:**

The authors aimed to investigate how genetic and environmental factors influence the muscle insulin signaling network and its impact on metabolism. They utilized mass spectrometry-based phosphoproteomics to quantify phosphosites in skeletal muscle of genetically distinct mouse strains in different dietary environments, with and without insulin stimulation. The results showed that genetic background and diet both affected insulin signaling, with almost half of the insulin-regulated phosphoproteome being modified by genetic background on an ordinary diet, and high-fat high-sugar feeding affecting insulin signaling in a strain-dependent manner.

Strengths:

Study uses state-of-the-art phosphoproteomics workflow allowing quantification of a large number of phosphosites in skeletal muscle, providing a comprehensive view of the muscle insulin signaling network. The study examined five genetically distinct mouse strains in two dietary environments, allowing for the investigation of the impact of genetic and environmental factors on insulin signaling. The identification of coreregulated subnetworks

within the insulin signaling pathway expanded our understanding of its organization and provided insights into potential regulatory mechanisms. The study associated diverse signaling responses with insulin-stimulated glucose uptake, uncovering regulators of muscle insulin responsiveness.

Weaknesses:

The limitations acknowledged by the authors, such as the need for larger cohorts and the inclusion of female mice. Moreover as acknowledged by authors, they are unable to dissect to what extent the obesity and different life span cycle for different strain affects insulin signaling. This suggest that further research is needed to validate and expand upon the findings.

- <https://doi.org/10.7554/eLife.89212.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The authors focused on genetic variability in relation to insulin resistance. They used genetically different lines of mice and exposed them to the same diet. They found that genetic predisposition impacts the overall outcome of metabolic disturbances. This work provides a fundamental novel view on the role of genetics and insulin resistance.

Reviewer #2 (Public Review):

Summary:

In the present study, van Gerwen et al. perform deep phosphoproteomics on muscle from saline or insulin-injected mice from 5 distinct strains fed a chow or HF/HS diet. The authors follow these data by defining a variety of intriguing genetic, dietary, or gene-by-diet phospho-sites that respond to insulin accomplished through the application of correlation analyses, linear mixed models, and a module-based approach (WGCNA). These findings are supported by validation experiments by intersecting results with a previous profile of insulin-responsive sites (Humphrey et al, 2013) and importantly, mechanistic validation of Pfkfb3 where overexpression in L6 myotubes was sufficient to alter fatty acid-induced impairments in insulin-stimulated glucose uptake. To my knowledge, this resource provides the most comprehensive quantification of muscle phospho-proteins which occur as a result of diet in strains of mice where genetic and dietary effects can be quantifiably attributed in an accurate manner. Utilization of this resource is strongly supported by the analyses provided highlighting the complexity of insulin signaling in muscle, exemplified by contrasts to the "classically-used" C57BL6/J strain. As it stands, I view this exceptional resource as comprehensive with compelling strength of evidence behind the mechanism explored. Therefore, most of my comments stem from curiosity about pathways within this resource, many of which are likely well beyond the scope of incorporation in the current manuscript. These include the integration of previous studies investigating these strains for changes in transcriptional or proteomic profiles and intersections with available human phospho-protein data, many of which have been generated by this group.

Strengths:

Generation of a novel resource to explore genetic and dietary interactions influencing the phospho-proteome in muscle. This is accompanied by the elegant application of in silico tools to highlight the utility.

Weaknesses:

Some specific aspects of integration with other data among the same fixed strains could be strengthened and/or discussed.

Reviewer #3 (Public Review):

Summary:

The authors aimed to investigate how genetic and environmental factors influence the muscle insulin signaling network and its impact on metabolism. They utilized mass spectrometry-based phosphoproteomics to quantify phosphosites in the skeletal muscle of genetically distinct mouse strains in different dietary environments, with and without insulin stimulation. The results showed that genetic background and diet both affected insulin signaling, with almost half of the insulin-regulated phosphoproteome being modified by genetic background on an ordinary diet, and high-fat high-sugar feeding affecting insulin signaling in a strain-dependent manner.

Strengths:

The study uses state-of-the-art phosphoproteomics workflow allowing quantification of a large number of phosphosites in skeletal muscle, providing a comprehensive view of the muscle insulin signaling network. The study examined five genetically distinct mouse strains in two dietary environments, allowing for the investigation of the impact of genetic and environmental factors on insulin signaling. The identification of coregulated subnetworks within the insulin signaling pathway expanded our understanding of its organization and provided insights into potential regulatory mechanisms. The study associated diverse signaling responses with insulin-stimulated glucose uptake, uncovering regulators of muscle insulin responsiveness.

Weaknesses:

Different mouse strains have huge differences in body weight on normal and high-fat high-sugar diets, which makes comparison between the models challenging. The proteome of muscle across different strains is bound to be different but the changes in protein abundance on phosphosite changes were not assessed. Authors do get around this by calculating 'insulin response' because short insulin treatment should not affect protein abundance. The limitations acknowledged by the authors, such as the need for larger cohorts and the inclusion of female mice, suggest that further research is needed to validate and expand upon the findings.

Reviewer #1 (Recommendations For The Authors):

I would suggest further discussion of the potential differences between males and females of the various strains.

In the revised manuscript we have included a more detailed discussion of the potential differences between male and female mice in the "Limitations of this study" section on lines 455-459. In particular, a landmark study of HFD-fed inbred mouse strains found that insulin sensitivity, as inferred from the proxy HOMA-IR, was affected by interactions between sex and strain despite generally being greater in female mice (10.1016/j.cmet.2015.01.002). Furthermore, a recent phosphoproteomics study of human induced pluripotent stem-cell derived myoblasts identified groups of insulin-regulated phosphosites affected by donor sex,

and by interactions between sex and donor insulin sensitivity (10.1172/JCI151818). Based on these results, we anticipate that both soleus insulin sensitivity and phosphoproteomic insulin responses would differ between male and female mice through interactions with strain and diet, adding yet another layer of complexity to what we observed in this study. This will be an important avenue for future research to explore.

Reviewer #2 (Recommendations For The Authors):

The following are comments to authors - many, if not all are suggestions for extended discussion and beyond the scope of the current elegant study.

In the discussion section (line 428) the authors make a key point in that the genetic, dietary, and interacting patterns of variation of Phospho-sites could be due to changes in total protein and/or transcript levels across strains. For example, given the increased expression of Pfkfb3 was sufficient to impact glucose uptake, suggesting that the transcript levels of the gene might also show a similar correlation with insulin responsiveness as in Fig 6b. Undoubtedly, phospho-proteomics analyses will provide unique information on top of more classical omics layers and uncover what would be an important future direction. Therefore, I would suggest adding to the discussion some guidance on performing similar applications to datasets from, at least some, of the strains used where RNA-seq and proteomics are available.

We thank the reviewer for this suggestion. To address this, we mined recently published total proteomics data collected from soleus muscles of seven CHOW or HFD-fed inbred mouse strains, three of which were in common with our study (C57Bl6J, BXH9, BXD34; 10.1016/j.cmet.2021.12.013). In this study ex vivo soleus glucose uptake was measured and correlation analysis was performed, so we directly extracted the resulting glucose uptake-protein associations and compared them to the glucose uptake-phosphoprotein associations identified in our study. Indeed, we found that only a minority of proteins correlated at both the phosphosite and total protein levels, highlighting the utility of phosphoproteomics to provide orthogonal information to more classical omics layers. We have included this analysis in lines 303-311.

Relevant to this, the authors might want to consider depositing scripts to analyze some aspects of the data (ex. WGCNA on P-protein data or insulin-regulated anova) in a repository such as github so that these can be applied easily to other datasets.

We refer the reviewer to the section "Code availability" on lines 511-513, where we deposited all code used to analyse the data on github.

In contrast to the points above, I feel that the short time-course of insulin stimulation was one important aspect of the experimental design that was not emphasized enough as a strength. It was mentioned as a limitation in that other time points could provide more info, yes. But given that the total abundance of proteins and transcripts likely doesn't shift tremendously in this time frame, this provides an important appeal to the analysis of phosphor-proteomic data. I would suggest highlighting the insulin-stimulated response analysis here as something that leverages the unique nature of phosphoproteomics.

We are grateful for the reviewer's positivity regarding this aspect of our experimental design. We have reiterated the value of the 10min insulin stimulation - that it temporally segregates phosphoproteomic and total proteomic changes - in the "Limitations of this study" section on lines 477-481.

While I recognize the WGCNA analysis as an instrumental way to highlight global patterns of phospho-peptide abundance co-regulation, the analysis currently seems somewhat underdeveloped. For example, Fig 5f-h shows a lot of overlap between kinase substrates and pathways among modules. Clearly, there are informative differences based on the intersection with Humphries 2013 and the correlation with Pfkfb3. To highlight the specific membership of these modules, most people rank-order module members by correlation with eigen-gene (or P-peptide) and then perform pathway enrichments on these. Alternatively, it looks like all data was used to generate modules across conditions. One consideration would be to perform WGCNA on relevant comparison data separately (ex. chow mice only and HFHS only) and then compare modules whose membership is retained or shift between the two. Or even look at module representation for genes that show large correlations with insulin-responsiveness. This might also be a good opportunity to suggest readers intersect module members with muscle eQTLs which colocalize to glucose or insulin to prioritize some potential key drivers.

We thank the reviewer for their helpful suggestions, which we feel have substantially improved the WGCNA analysis. To probe specific functional differences between subnetworks, we performed rank-based enrichment using phosphopeptide module membership scores. Interestingly, this did reveal pathways that were enriched only in certain modules. However, we found that after p-value adjustment, virtually all enriched pathways lost statistical significance, hence we interpret these results as suggestive only. We have made this analysis available to readers in Fig S4b-d and lines 263-265: "To further probe functional differences we analysed phosphopeptide subnetwork membership scores, which revealed additional pathways enriched in individual subnetworks. However, these results were not significant after p-value adjustment and hence are suggestive only (Fig. S4b-d)". We also visualised module representation for glucose-uptake correlated phosphopeptides. This agreed with our existing analysis in Fig. 6f, where the eigenpeptides of modules V and I were correlated with glucose uptake (Fig. 6f). We have incorporated this new analysis in Fig. S6b-c and lines 324-325: "Examining the subnetwork membership scores for glucose-uptake correlated phosphopeptides also revealed a preference for clusters V and I, supporting this analysis (Fig. S6b-c)." Finally, in the discussion we have presented the integration of genetic data, such as muscle-specific eQTLs, as a future direction (lines 398-401): "Alternatively, one could overlap subnetworks with genetic information, such as genes associated with glucose homeostasis and other metabolic traits in human GWAS studies, or muscle-specific eQTLs or pQTLs genetically colocalised with similar traits, to further prioritise subnetwork-associated phenotypes and identify potential drivers within subnetworks."

Have the authors considered using their heritability and GxE estimated for module eigenpeptides? To my knowledge, this has never been performed and might provide some informative information as the co-regulated P-protein structure occurs as a result of relevant contexts.

In the revised manuscript we have now analysed eigenpeptides with the same statistical tests used to identify Strain and Diet effects in insulin-regulated phosphopeptides. We have displayed the statistical results in Fig. S4a, and have explicitly mentioned examples of StrainxDiet effects on lines 245-247: "For example, HFD-feeding attenuated the insulin response of subnetwork I in CAST and C57Bl6J strains (t-test adjusted $p = 0.0256, 0.0365$), while subnetwork II was affected by HFD-feeding only in CAST and NOD (Fig. 5e, Fig. S4a, t-test adjusted $p = 0.00258, 0.0256$)."

The integration of modules with adipocyte phosphoproteomic data from the authors 2013 Cell metab paper seems like an important way to highlight the integration of this

resource to define critical cellular signaling mechanisms. To assess the conservation of signaling mechanisms and relationships to additional key contexts (ex. exercise), the intersection of the insulin-stimulated P-peptides with human datasets generated by this group (ex. cell metab 2015, nature biotech 2022) seems like an obvious future direction to prioritize targets. Figure S3B shows a starting point for these types of integrations.

To demonstrate the value of integrating our results with related phosphoproteomics data, we have incorporated the reviewer's advice of comparing insulin-regulated phosphosites to exercise-regulated phosphosites from Needham et. Nature Biotech 2022 and Hoffman et al. Cell Metabolism 2015. We identified a small subset of commonly regulated phosphosites (8 across all three studies). Given insulin and exercise both promote GLUT4 translocation, these sites may represent conserved regulatory mechanisms. This analysis is presented in Fig. S3d, Table S2, and lines 129-135: "In addition to insulin, exercise also promotes GLUT4 translocation in skeletal muscle. We identified a small subset of phosphosites regulated by insulin in this study that were also regulated by exercise in two separate human phosphoproteomics studies (Fig. S3d, Table S2, phosphosites: Eef2 T57 and T59, Mff S129 and S131, Larp1 S498, Tbc1d4 S324, Svitl S300, Gys1 S645), providing a starting point for exploring conserved signalling regulators of GLUT4 translocation."

For the Pfkfb3 overexpression system, are there specific P-peptides that are increased/decreased upon insulin stimulation? This might be an interesting future direction to mention in order to link signaling mechanisms.

We assessed whether canonical insulin signalling was affected by Pfkfb3 overexpression by immunoblotting. Insulin-stimulated phosphorylation of Akt S473, Akt T308, Gsk3a/b S21/S9, and PRAS40 T246 differed little across conditions, with only a weak, statistically insignificant trend towards increased pT308 Akt, pS21/S9 Gsk3a/b, and pT246 PRAS40 in palmitate-treated Pfkfb3-overexpressing cells. Hence, as the reviewer has suggested, an interesting future direction will be to perform phosphoproteomics to characterise more deeply the effects of palmitate and Pfkfb3 overexpression on insulin signalling. We have modified the manuscript to reflect these findings and suggested future directions on lines 362-365: "immunoblotting of canonical insulin-responsive phosphosites on Akt and its substrates GSK3 α/β and PRAS40 revealed minimal effect of palmitate treatment and Pfkfb3 overexpression (Fig. S7e-f), hence more detailed phosphoproteomics studies are needed to clarify whether Pfkfb3 overexpression restored insulin action by modulating insulin signalling."

Reviewer #3 (Recommendations For The Authors):

This remarkable contribution by the esteemed research group has significantly enriched the field of metabolism. The extensive dataset, intertwined with a sophisticated research design, promises to serve as an invaluable resource for the scientific community. I offer a series of suggestions aimed at potentially elevating the manuscript to an even higher standard.

Mouse Weight Variation and Correlation Analysis: The pronounced variances in mouse body weights pose a challenge to meaningful comparisons (Fig S1). Could the disparities in the phosphoproteome between basal and insulin-stimulated conditions be attributed to differences in body weight? Consider performing a correlation analysis. Furthermore, does the phosphoproteome of these mouse strains evolve comparably over time? Do these mice age similarly? Kindly incorporate this information.

We thank the reviewer for the suggested analysis. We found there was a significant correlation between the phosphopeptide insulin response and mouse body weight, either in CHOW-fed mice (Strain effects) or across both diets (Diet effects), for ~ 25% of phosphopeptides that exhibited a Strain or Diet effect. Hence, while there is a clear effect of

body weight on insulin signalling, this influences only a small proportion of the entire insulin-responsive phosphoproteome. Notably, insulin was dosed according to mouse lean mass to ensure equivalent dosage received by the soleus muscle, hence any insulin signalling differences associated with body weight are unlikely due to differences in dosing. As the reviewer also alludes to, different strains could have different lifespans. This may result in mice having different biological ages at the time of experimentation, and this in turn could influence insulin signalling. This possibility is challenging to assess in a quantitative manner because lifespan data is not available for most strains used. However, it is worth noting that female CAST mice live 77% as long as C57Bl6J mice (median age of 671 vs 866 (10.1073/pnas.1121113109); data is not available for male mice nor the other three strains), and substantial differences in insulin signalling were observed between these two strains. Ultimately, regardless of whether body weight and/or lifespan altered insulin signalling, such differences would still have arisen solely from the distinct genetic backgrounds and diets of the mice, hence we believe they are meaningful results that should not be dismissed. We have added this analysis to the revised manuscript in the "Limitations of this study" section on lines 471-477: "We were also unable to determine the extent to which signalling changes arose from muscle-intrinsic or extrinsic factors. For instance, body weight varied substantially across mice and correlated significantly with 25% of Strain and Diet-affected phosphopeptides (Fig. S8c), suggesting obesity-related systemic factors likely impact a subset of the muscle insulin signalling network. Furthermore, genetic differences in lifespan could alter the "biological age" of different strains and their phosphoproteomes, though we could not assess this possibility since lifespan data are not available for most strains used. "

Soleus Muscle Data and Bias Considerations: Were measurements taken for lean mass and soleus muscle weight? If so, please present the corresponding data.

Measurements for lean mass and the mass of soleus muscle after grinding have been including in Supplementary Figure S1 (panels c-d)

As outlined in the methods section, the variation in protein yield from the soleus muscle across each strain is substantial. Notably, the distinct peptide input for phospho enrichment introduces biases, given that muscles with lower input may exhibit reduced identification (Fig S2). This bias might also manifest in the PCA plot (S2C). Ideally, adopting a uniform protein/peptide input would have been advantageous. Address this concern and contemplate moving the PCA plot to the main figure. It's prudent to reconsider the sentence stating, "Samples from animals of the same strain and diet were highly correlated and generally clustered together, implying the data are highly reproducible (Fig. S2b-d)," particularly if the input and total IDs were not matched.

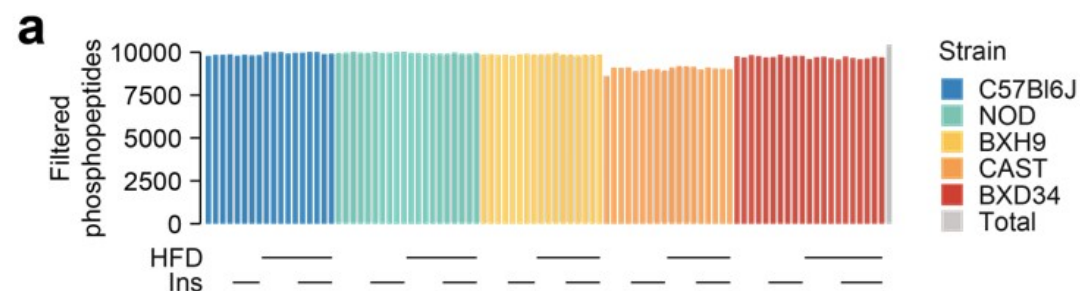
The reviewer highlights an important point. As the reviewer comments, it would have been our preference to use the same amount of protein material for all samples. However, as there was a wide range in the mass of the soleus muscle across mouse strains (in particular much lower in CAST mice), it was not appropriate to use the same amount of material for all strains. This is indeed evident in the PCA plot (Figure S2c), whereby samples cluster in the second component (PC2) based on the amount of protein material. However, this clustering is not observed in the hierarchical clustering (Figure S2d), and nor are the number of phosphopeptides quantified in each sample substantially impacted by these differences (Figure S2a) as implied by the reviewer. Indeed, the number of phosphopeptides quantified did not noticeably vary when comparing BXH9/BXD34 to C57Bl6J/NOD despite 32.3% less material used, and there were only 12.4% fewer phosphopeptides (average #13891.56 vs 15851.29) in CAST compared to C57Bl6J/NOD strains, despite 51.8% less material used. To further emphasise the minimal effect that input material had on phosphopeptide quantification, we have additionally plotted the number of phosphopeptides quantified in each sample following the filtering steps we employed prior to statistical analysis of the

dataset (i.e. ANOVA). This plot (Author response image 1) shows that there is even less variation in the number of quantified phosphopeptides between strains, with only 9.12% fewer phosphopeptides quantified and filtered on average in CAST compared to C57Bl6J/NOD (average #9026.722 vs 9932.711). From a quantitative perspective, in both the PCA (Principal Component 1) and hierarchical clustering analyses, samples are additionally clustered by individual strains, and in the latter they also cluster generally by diet, implying that biological variation between samples remains the primary variation captured in our data. We have modified the manuscript so that these observations are forefront (lines 103-106):

"Furthermore, while different strains clustered by the amount of protein material used in the second component of the PCA (Figure S2c), samples from animals of the same strain and diet were highly correlated and generally clustered together, indicating that our data are highly reproducible". To ensure that readers are aware of our decision to alter protein starting material and its implications, we have moved the description of this from the methods to the results, and we have highlighted the impact on phosphopeptide quantification in CAST mice (lines 99-103): "Due to the range in soleus mass across strains (Fig. S1D) we altered the protein material used for EasyPhos (C57Bl6J and NOD: 755 µg, BXH9 and BXD34: 511 µg, CAST: 364 µg), though phosphopeptide quantification was minimally affected, with only 12.4% fewer phosphopeptides quantified on average in CAST compared to the C57Bl6J/NOD (average 13891.56 vs 15851.29 Fig. S2a)."

Author response image 1.

Phosphopeptide quantification following filtering. a) The number of phosphopeptides quantified in each sample after filtering prior to statistical analysis.



Phosphosite Quantification Filtering: The quantified phosphosites have been dropped from 23,000 to 10,000. Could you elucidate the criteria employed for filtering and provide a concise explanation in the main text?

We thank the reviewer for drawing this ambiguity to our attention. Before testing for insulin regulation, we performed a filtering step requiring phosphopeptides to be quantified well enough for comparisons across strains and diets. Specifically, phosphopeptides were retained if they were quantified well enough to assess the effect of insulin in more than eight strain-diet combinations (≥ 3 insulin-stimulated values and ≥ 3 unstimulated values in each combination). We have now included this explanation of the filtering in the main text on lines 108-114.

ANOVA Choice Clarification: In Figure 4, there's a transition from one-way ANOVA in B to two-way ANOVA in C. Could you expound on the rationale for selecting these distinct methods?

In panel B, we first focussed on kinase regulation differences between strains in the absence of a dietary perturbation. Hence, we performed one-way ANOVAs only within the CHOW-fed

mice. In panel C, we then consider the effect of perturbation with the HFD. We perform two-way ANOVAs, allowing us to identify effects of the HFD that are uniform across strains (Diet main effect) or variable across strains (Strain-by-diet interaction).

Cell Line Selection for Functional Experiments: Could you elucidate the rationale behind opting for L6 cells of rat origin over C2C12 mouse cells for functional experiments?

We acknowledge that C2C12 cells have the benefit of being of mouse origin, which aligns with our mouse-derived phosphoproteomics data. However, they are unsuitable for glucose uptake experiments as they lack an insulin-responsive vesicular compartment even upon GLUT4 overexpression, and undergo spontaneous contraction when differentiated resulting in confounding non-insulin dependent glucose uptake (10.1152/ajpendo.00092.2002, 10.1007/s11626-999-0030-8). In contrast, L6 cells readily express insulin-responsive GLUT4, and cannot contract (doi.org/10.1113/JP281352, 10.1007/s11626-999-0030-8). Therefore they are a superior model for studying insulin-dependent glucose transport. We have added a justification of L6 cells over C2C12 cells in the revised manuscript, on lines 352-354: "While L6 cells are of rat origin, they are preferable to the popular C2C12 mouse cell line since the latter lack an insulin-responsive vesicular compartment and undergo spontaneous contraction, resulting in confounding non-insulin dependent glucose uptake."

It's intriguing that while a phosphosite was modulated on Pfkfb2, functional assays were conducted on a different isoform (Pfkfb3) wherein the phosphosite was not detected.

The correlation between Pfkfb2 S469 phosphorylation and insulin-stimulated glucose uptake suggests that F2,6BP production, and subsequent glycolytic activation, positively regulate insulin responsiveness. There are several ways of testing this: 1) Knock down endogenous Pfkfb2, and re-express either wild-type protein or a S469A phosphomutant. If S469 phosphorylation positively regulates insulin responsiveness, then knockdown should decrease insulin responsiveness and re-expression of wild-type Pfkfb2, but not S469A, should restore it. 2) Induce insulin resistance (e.g. through palmitate treatment), and overexpress phosphomimetic S469D or S469E Pfkfb2 to enhance F2,6BP production. Under our hypothesis, this should reverse insulin resistance. 3) There is some evidence that dual phosphorylation of S469 and S486, another activating phosphosite on Pfkfb2, enhances F2,6BP production through 14-3-3 binding (10.1093/emboj/cdg363). Hence, we may expect that introduction of an R18 sequence into Pfkfb2, which causes constitutive 14-3-3 binding (10.1074/jbc.M603274200), would increase Pfkfb2-driven F2,6BP production, and under our hypothesis this should reverse insulin resistance. 4) The paralog Pfkfb3 lacks Akt regulatory sites and has substantially higher basal activity than Pfkfb2. Thus, overexpression of Pfkfb3 should mimic the effect of phosphorylated Pfkfb2, and hence reverse insulin resistance under our hypothesis. While approaches 1), 2), and 3) directly target Pfkfb2, they have drawbacks. For example, 1) may not work if Pfkfb2 knockdown is compensated for by other Pfkfb isoforms, 2) may not work since D/E phosphomimetics often do not recapitulate the molecular effects of S/T phosphorylation (10.1091/mbc.E12-09-0677), and 3) may not work if S469 phosphorylation does not operate through 14-3-3 binding. Hence we performed 4) as it seemed to be the most robust and cleanest experiment to test our hypothesis. We have revised the manuscript to further clarify the challenges of directly targeting Pfkfb2 and the benefits of targeting Pfkfb3 on lines 342-349: "Since Pfkfb2 requires phosphorylation by Akt to produce F2,6BP substantially, increasing F2,6BP production via Pfkfb2 would require enhanced activating site phosphorylation, which is difficult to achieve in a targeted fashion, or phosphomimetic mutation of activating sites to aspartate/glutamate, which often does not recapitulate the molecular effects of serine/threonine phosphorylation. By contrast, the paralog Pfkfb3 has high basal production rates and lacks an Akt motif at the corresponding phosphosites. We therefore rationalised that overexpressing Pfkfb3 would robustly increase F2,6BP production and enhance glycolysis regardless of insulin stimulation and Akt signalling."

Insulin-Independent Action of Pfkfb3: The functionality of Pfkfb3 unfolds in an insulin-independent manner, yet it restores insulin action (Fig 6h). Could you shed light on the mechanism underpinning this phenomenon? Consider measuring F2,6BP concentrations or assessing kinase activity upon overexpression.

Pfkfb3 overexpression increased the glycolytic capacity of L6 myotubes in the absence of insulin stimulation, as inferred by extracellular acidification rate (Fig. S7c). This is indeed consistent with Pfkfb3 enhancing glycolysis through increased F2,6BP concentration in an insulin-independent manner. To shed light on the mechanism connecting this to insulin action, we performed immunoblotting experiments to assess the kinase activity of Akt, a master regulator of the insulin response. Indeed, this experimental direction has precedent as we previously observed that Pfkfb3 overexpression enhanced insulin-stimulated Akt signalling in HEK293 cells, while small-molecule inhibition of Pfkfb kinase activity reduced Akt signalling in 3T3-L1 adipocytes (10.1074/jbc.M115.658815). However, insulin-stimulated phosphorylation of Akt S473, Akt T308, Gsk3a/b S21/S9, and PRAS40 T246 differed little across conditions, with only a weak, statistically insignificant trend towards increased pT308 Akt, pS21/S9 Gsk3a/b, and pT246 PRAS40 in palmitate-treated Pfkfb3-overexpressing cells. Hence, a more detailed phosphoproteomics study will be needed to assess whether Pfkfb3 restores insulin action by modulating insulin signalling. We have described these immunoblotting experiments in lines 361-365 and Fig. S7e-f. We also discussed potential mechanisms through which Pfkfb3-enhanced glycolysis could connect to insulin action in the discussion (lines 427-434).

Figure 6h Statistical Analysis: For the 2DG uptake in Figure 6h, a conventional two-way ANOVA might be more appropriate than a repeated measures ANOVA.

On reflection, we agree that a conventional ANOVA is more appropriate. Furthermore, for simplicity and conciseness we have decided to analyse and present only insulin-stimulated/unstimulated 2DG uptake fold change values in Figure 6h. We have presented all unstimulated and insulin-stimulated values in Figure S7d.